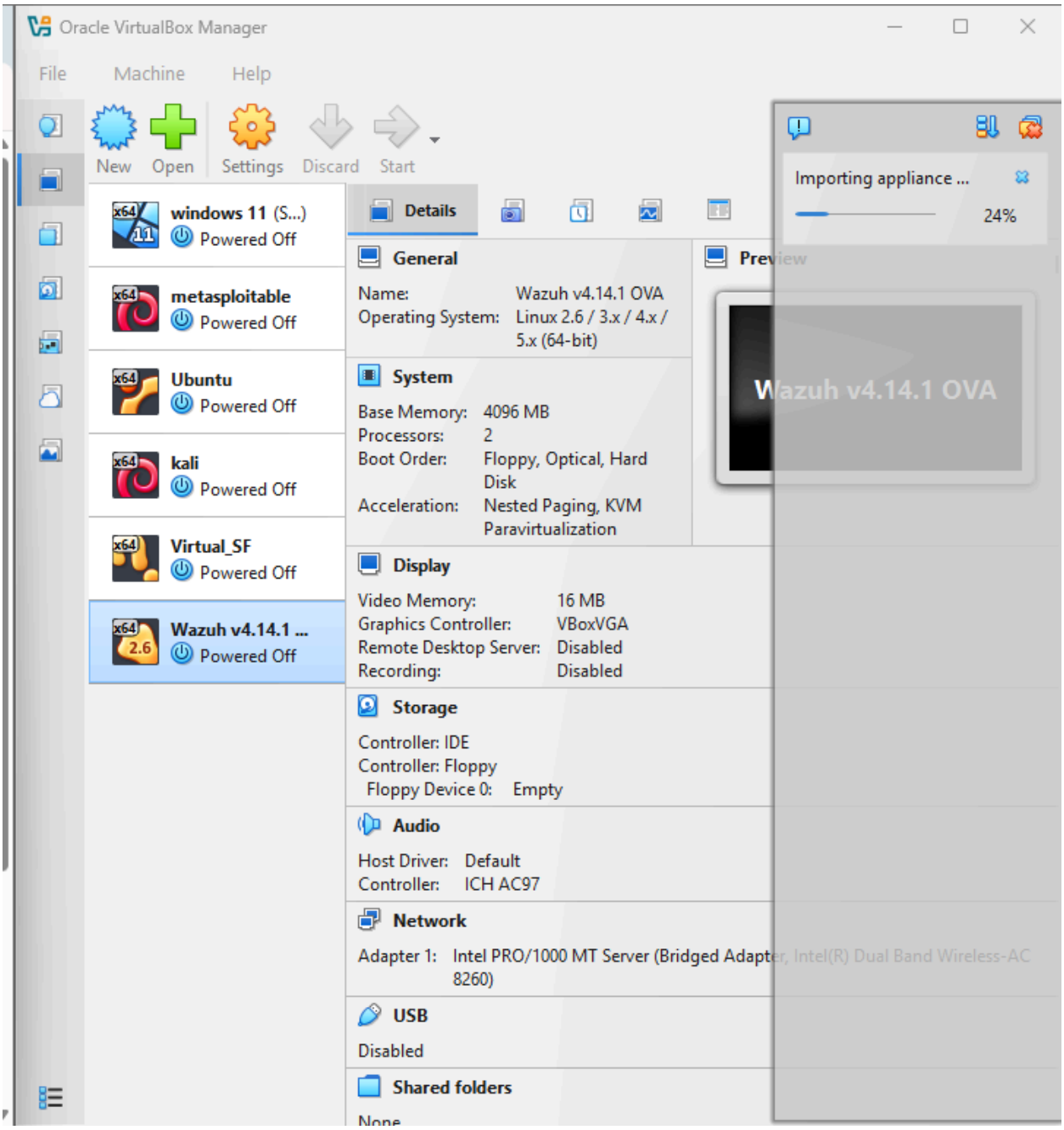
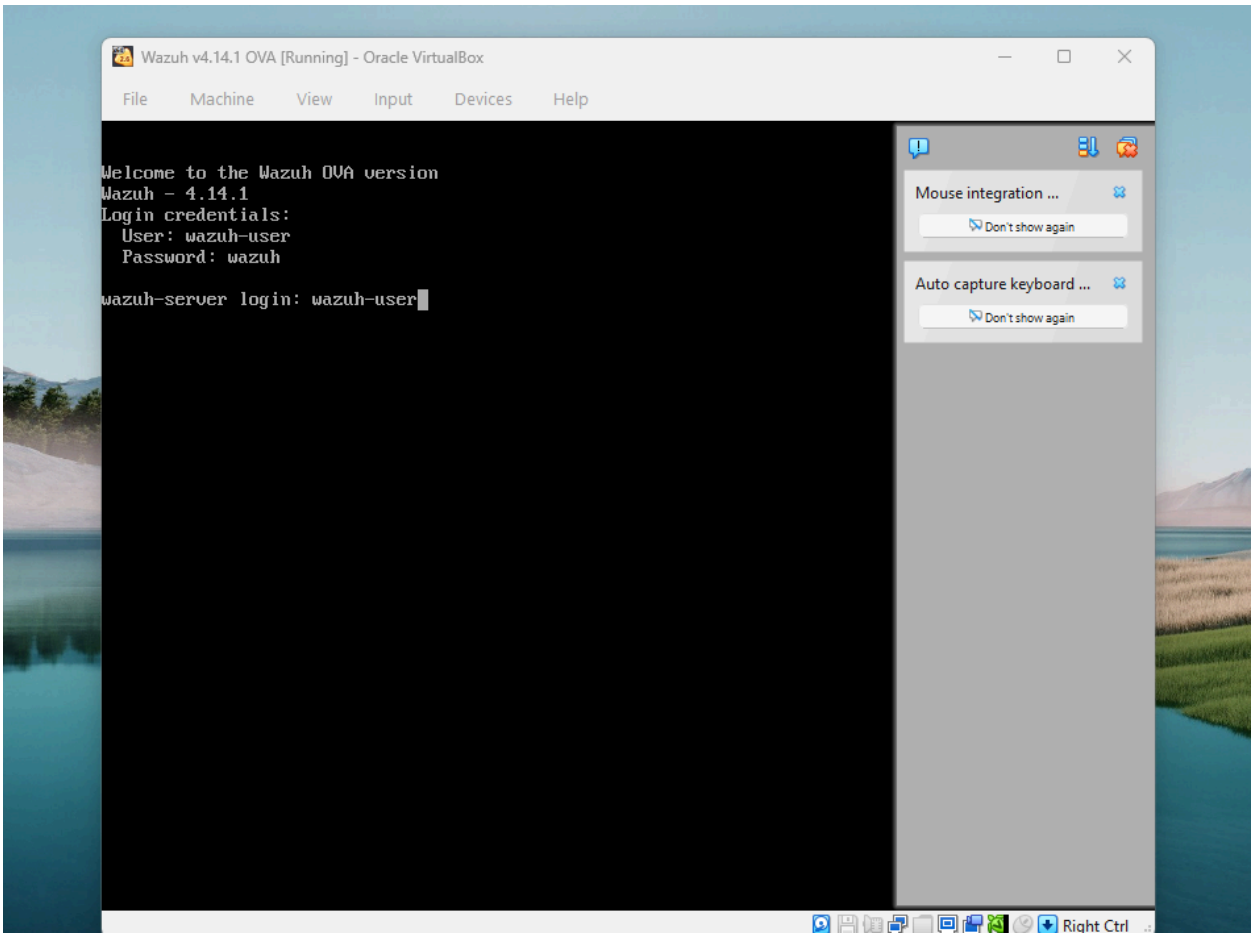
Wazuh Endpoint Configuration And Security Monitoring

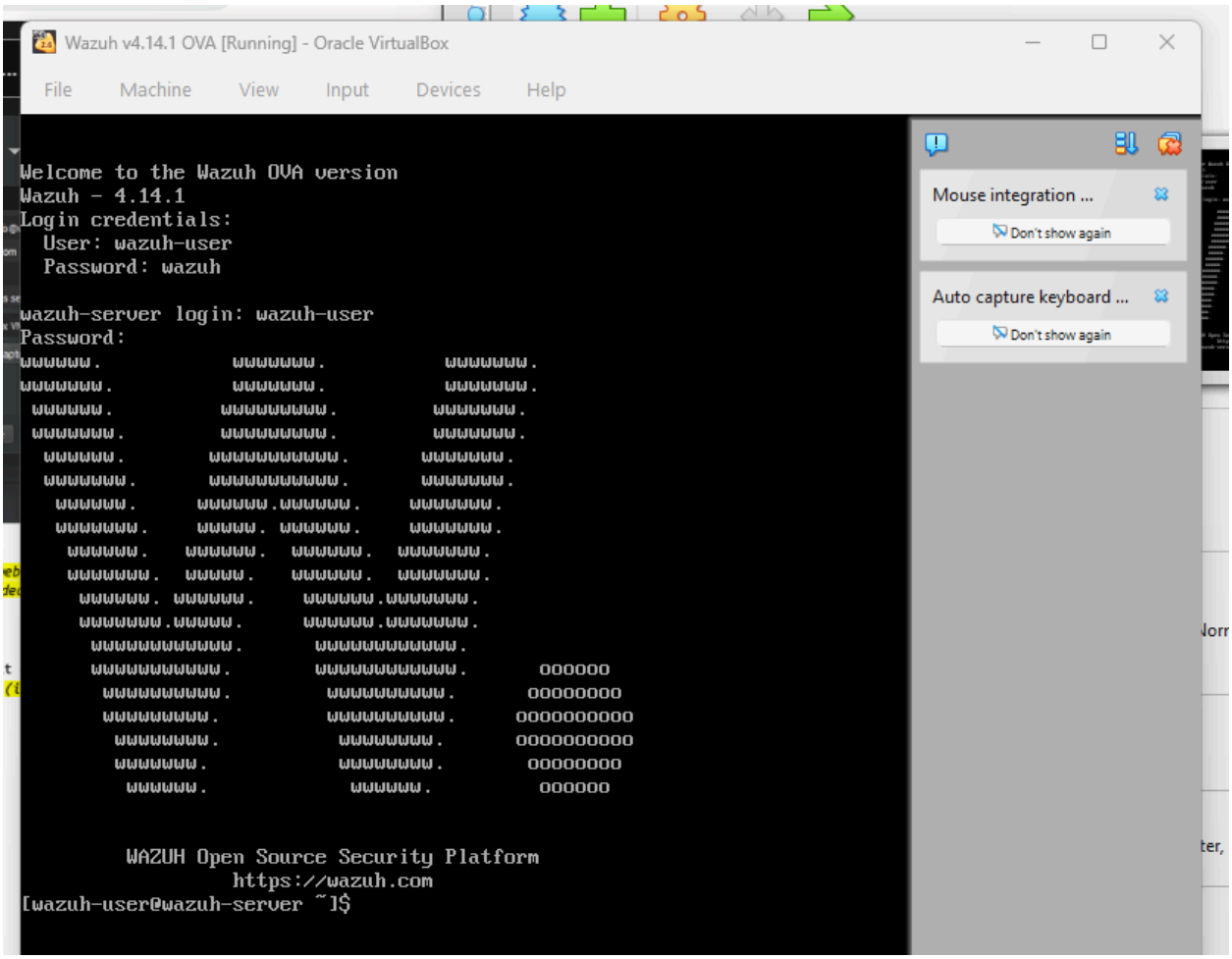
Wazuh Manager VM Setup on Windows Host & Endpoint
Policy Configuration

1. The Wazuh OVA was downloaded and imported into **Oracle VirtualBox**. The VM was provisioned with **4096 MB of RAM** and **2 Processors** to ensure stable performance.

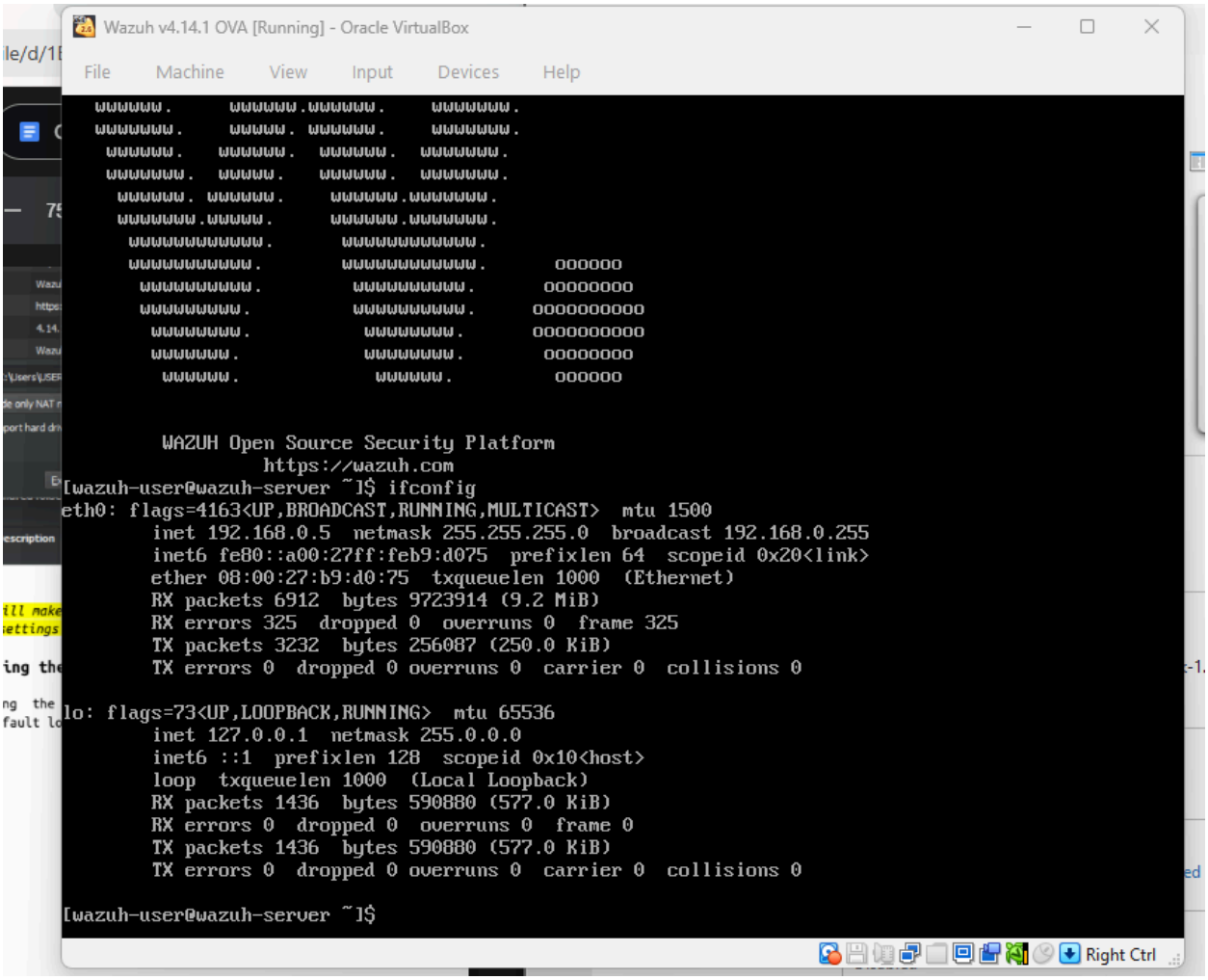


2. I Started the Wazuh VM and Logged in using the default credentials “USER: wazuh-user &
   Password: wazuh”

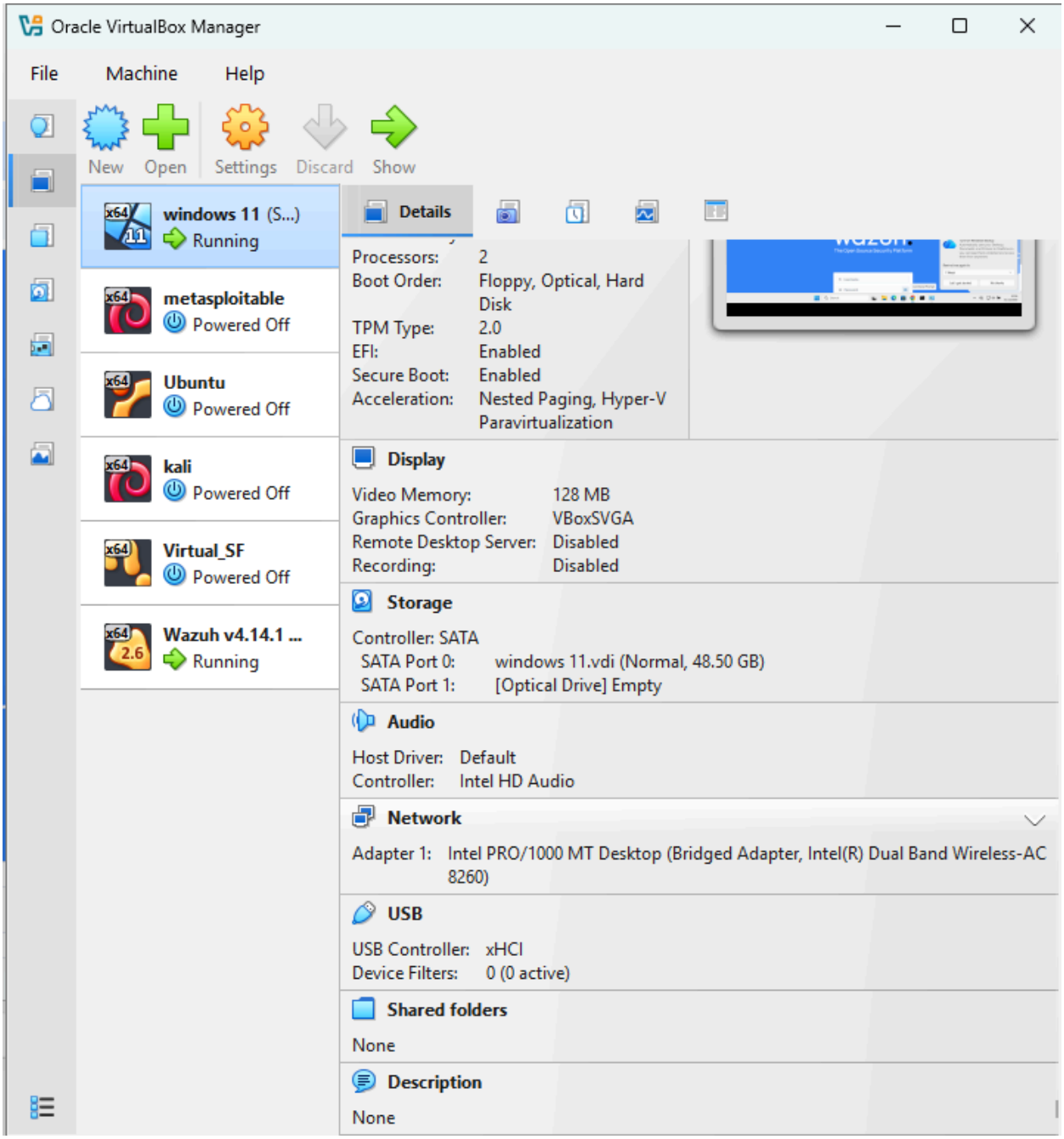

Wazuh v4.14.1 OVA [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Welcome to the Wazuh OVA version
Wazuh - 4.14.1
Login credentials:
User: wazuh-user
Password: wazuh
wazuh-server login: wazuh-user
Mouse integration ...
Don't show again
Auto capture keyboard ...
Don't show again
Right Ctrl



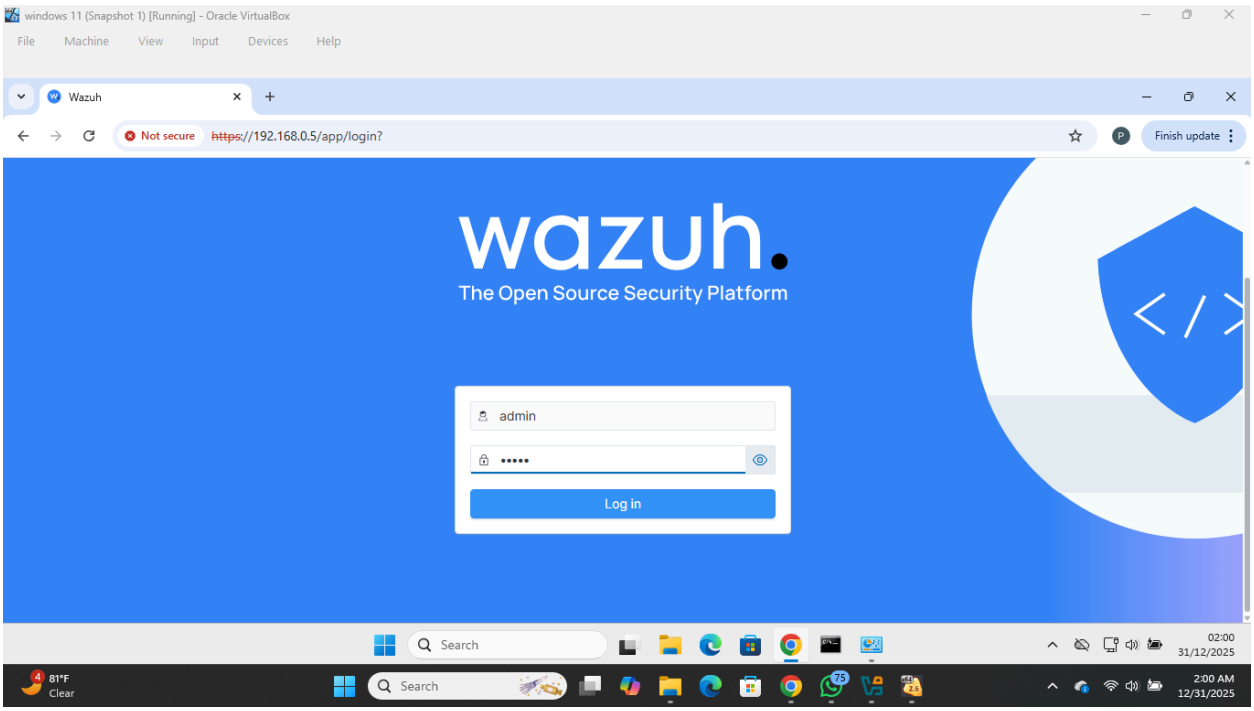
3. The `ifconfig` command was executed within the manager's CLI to identify its IP address, **192.168.0.5**. This IP is essential for establishing communication with endpoints and accessing the web-based dashboard.



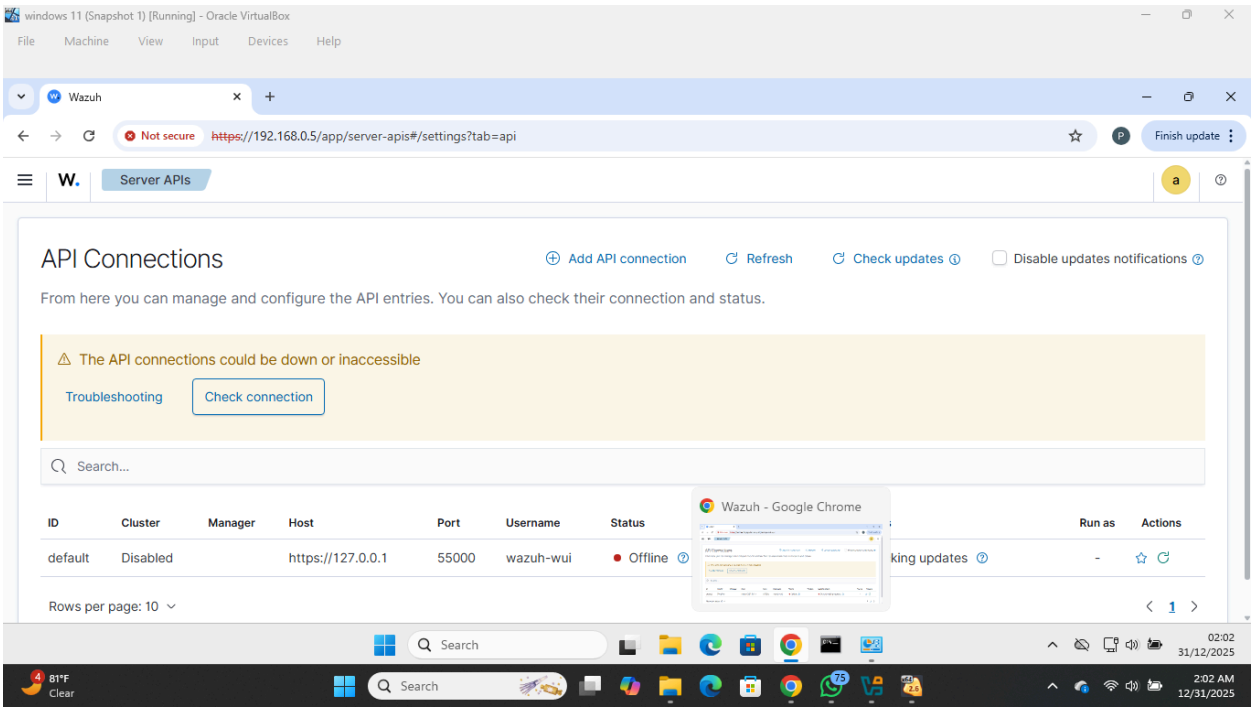
4. I will be using my windows virtual machine as my wazuh endpoint



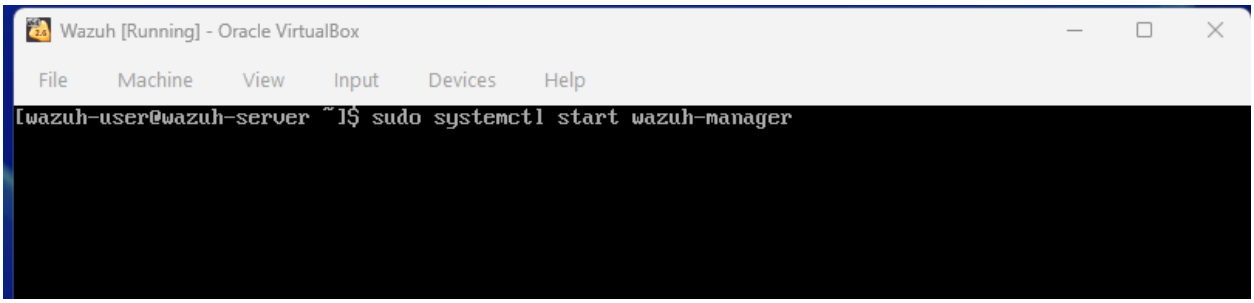
5. Using the wazuh's ip address, I logged into my admin dashboard



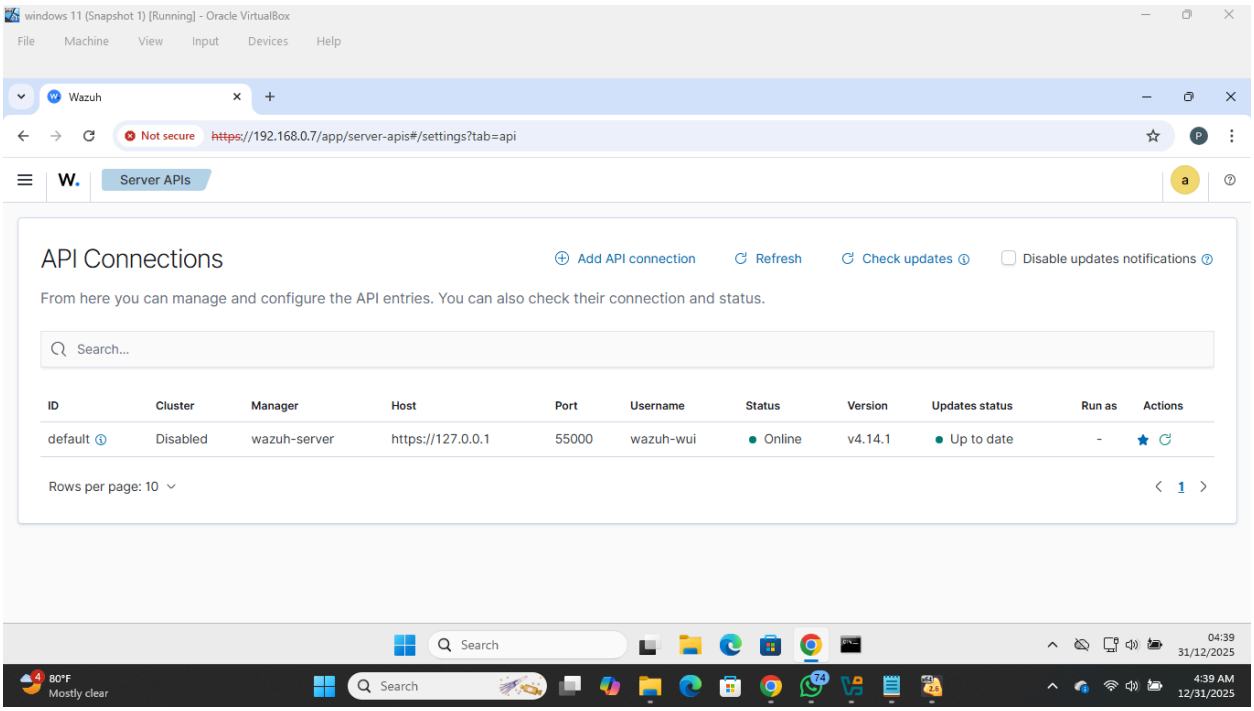
6. After attempting to access the dashboard through a web browser, an **API connection error** was encountered, showing a status of **Offline**.
   Root Cause: Troubleshooting revealed that the error stemmed from the `wazuh-manager` daemons failing to start properly during the initial boot sequence.



7. To resolve this, I restarted the wazuh-manager using the command `"sudo systemctl start wazuh-manager"`. This command properly starts the wazuh-manager and the error should be resolved.

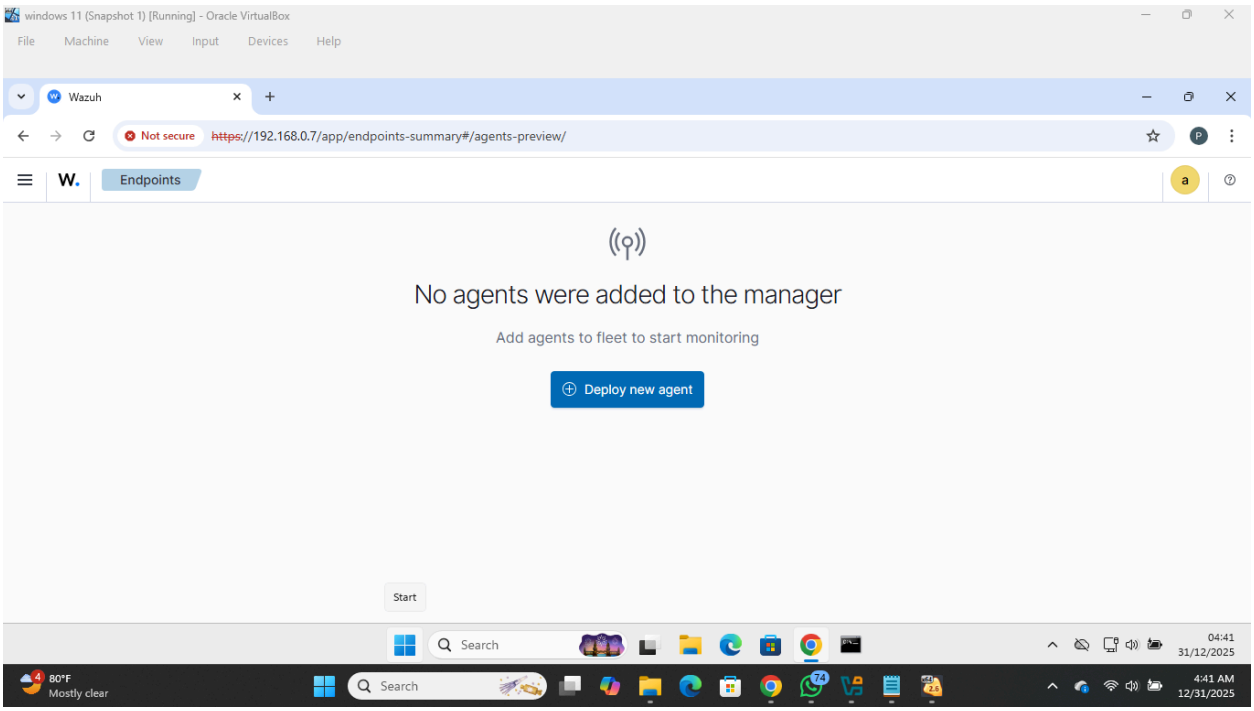


8. A page refresh confirmed the dashboard had successfully connected to the API with a status of **Online**

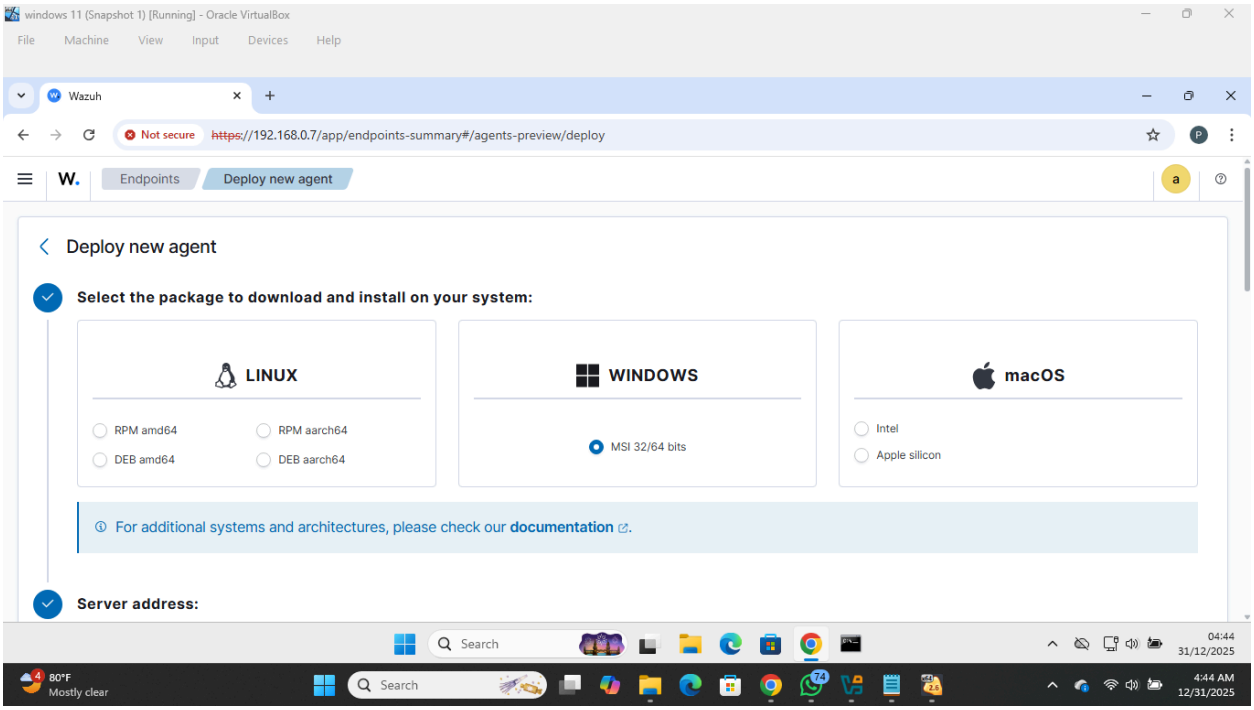


Agent Installation on the Windows Endpoint

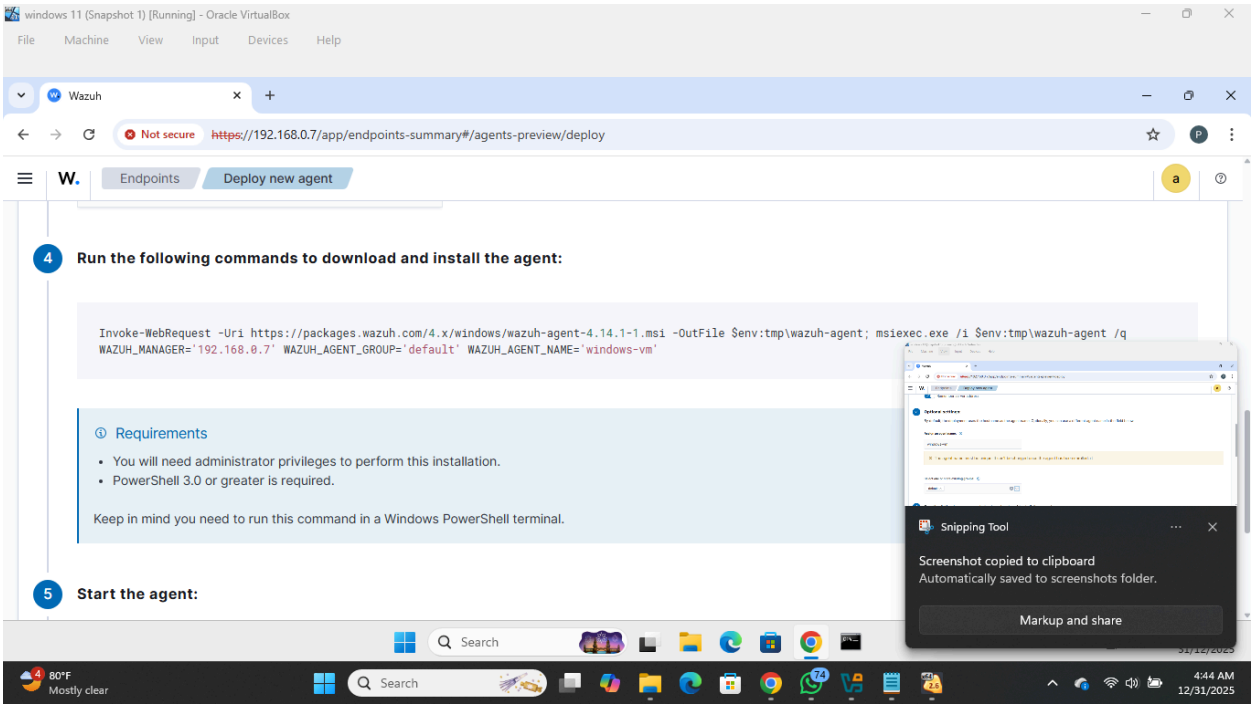
1. To add my windows endpoint, On the menu bar I clicked on “Agent Management” then “summary”.
   I clicked on the “Deploy new agent” option to add my windows Virtual machine to my wazuh.



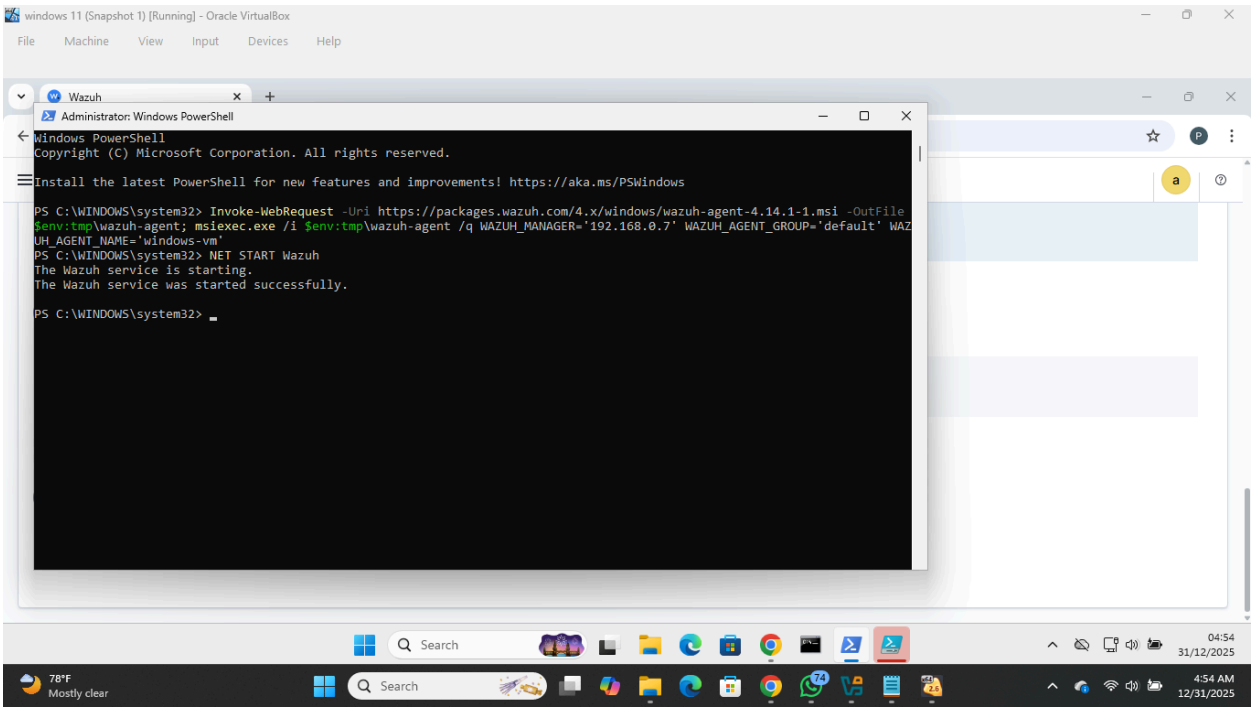
2. For my agent configuration, I am working with a windows machine and I will name this agent “windows-vm” .



3. Next, I copied the command as directed to run on my powershell

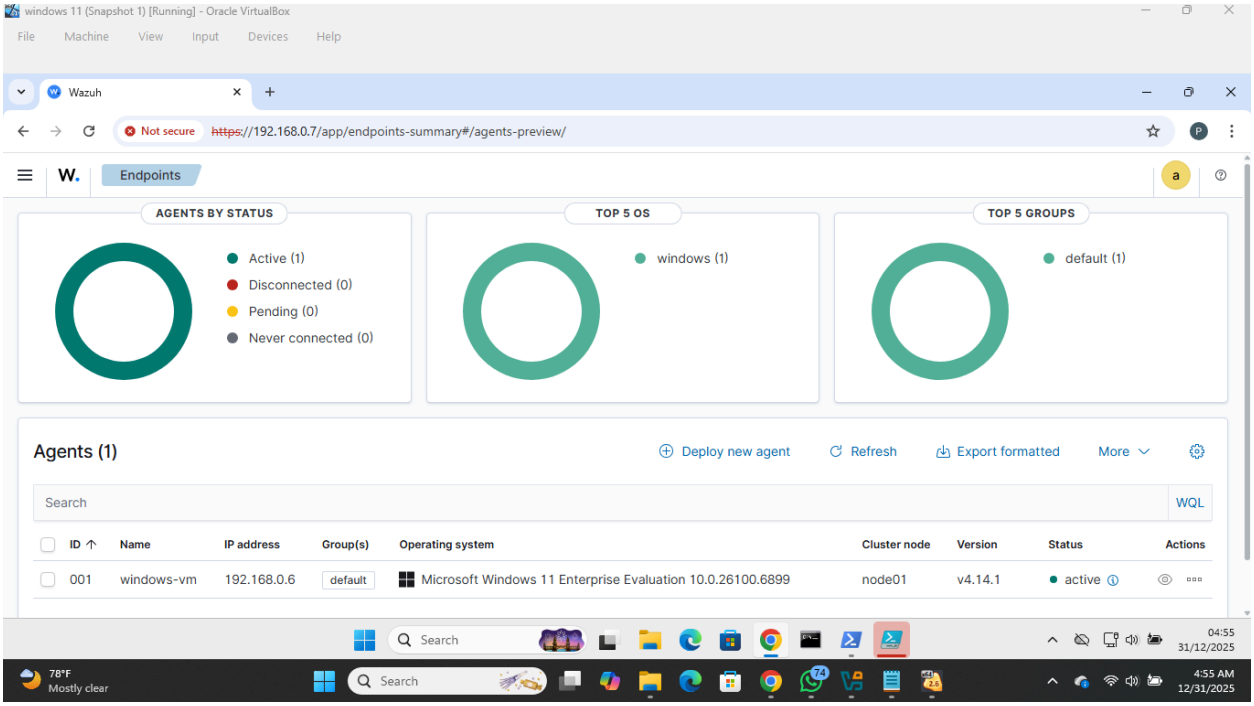


4. On the Windows VM (the endpoint), I opened my PowerShell as Administrator.
   Paste the long command I copied from the wazuh dashboard.
   Then copied the next command to start the agent



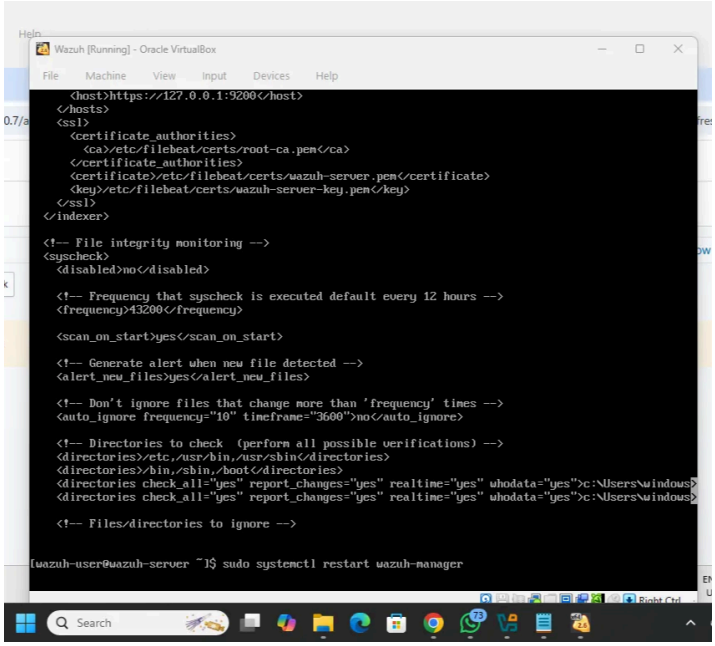
5. Verifying Agent Connection in the Dashboard

I Returned to my Wazuh Dashboard, Went to Agents Management and confirmed my windows vm was added as an agent.

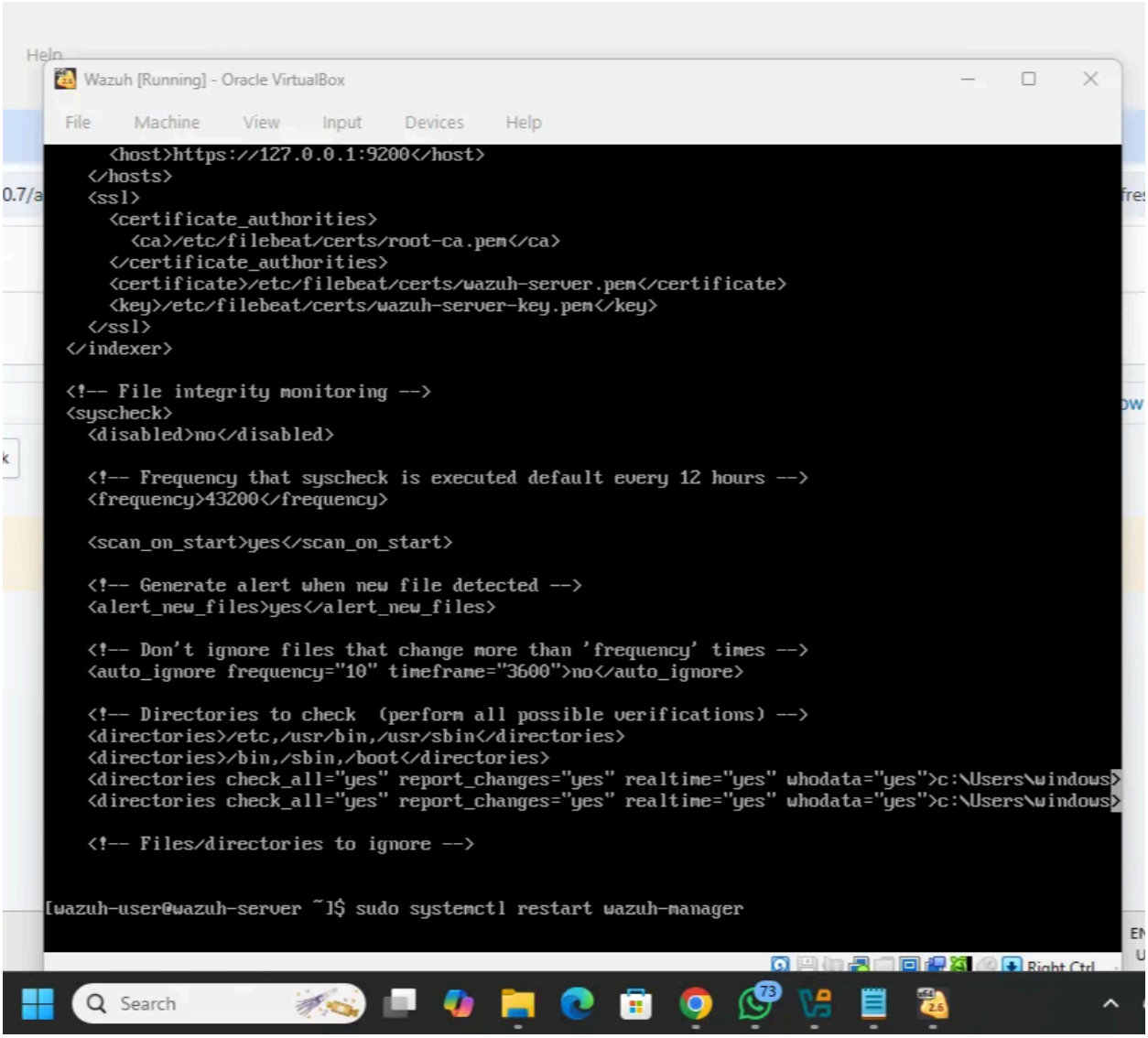


Policy Configuration - File Integrity Monitoring (FIM)

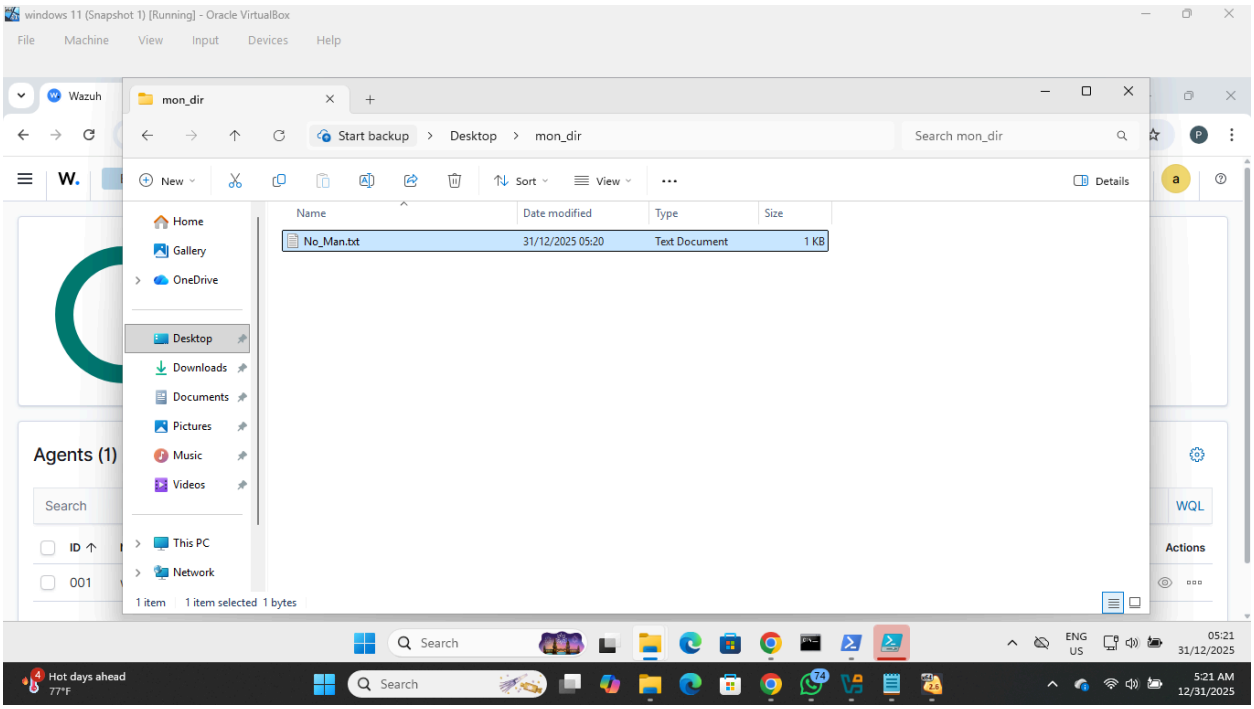
1. To enable File Integrity Monitoring on my windows VM, I will add the path to the file or directory I want to monitor to the “ossec.conf” file on my wazuh VM.
   On the Wazuh VM, I opened the main configuration file:
   ”nano /var/ossec/etc/ossec.conf”
   Under the <syscheck> block I added two new <directories> tags specifying the full path to the file and directory I want to monitor on my Windows laptop.
   “C:\Users\windows\Desktop\mon_dir\No_Man.txt”
   ”C:\Users\windows\Desktop\razor”



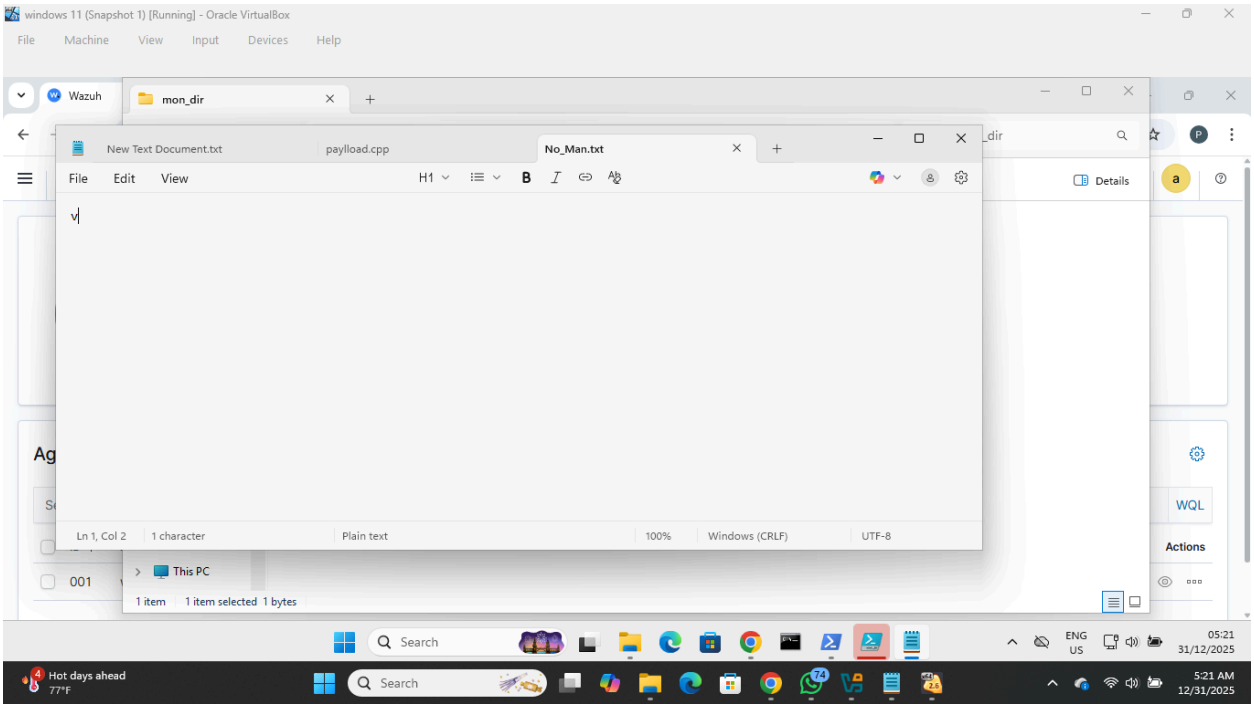
2. After adding the necessary configurations to the file, I saved it.
   Next, I restarted the wazuh-manager so the new configurations can take effect.



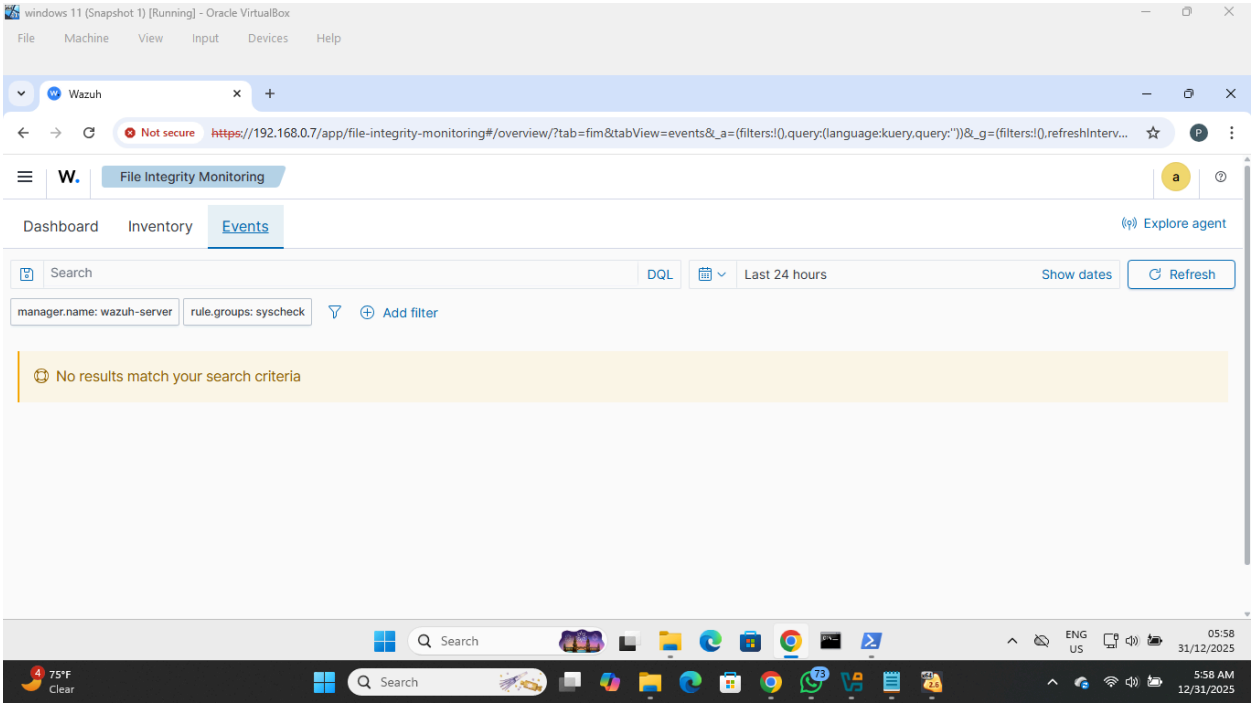
3. To test the FIM configuration, I modified the text file which was being monitored by the wazuh FIM



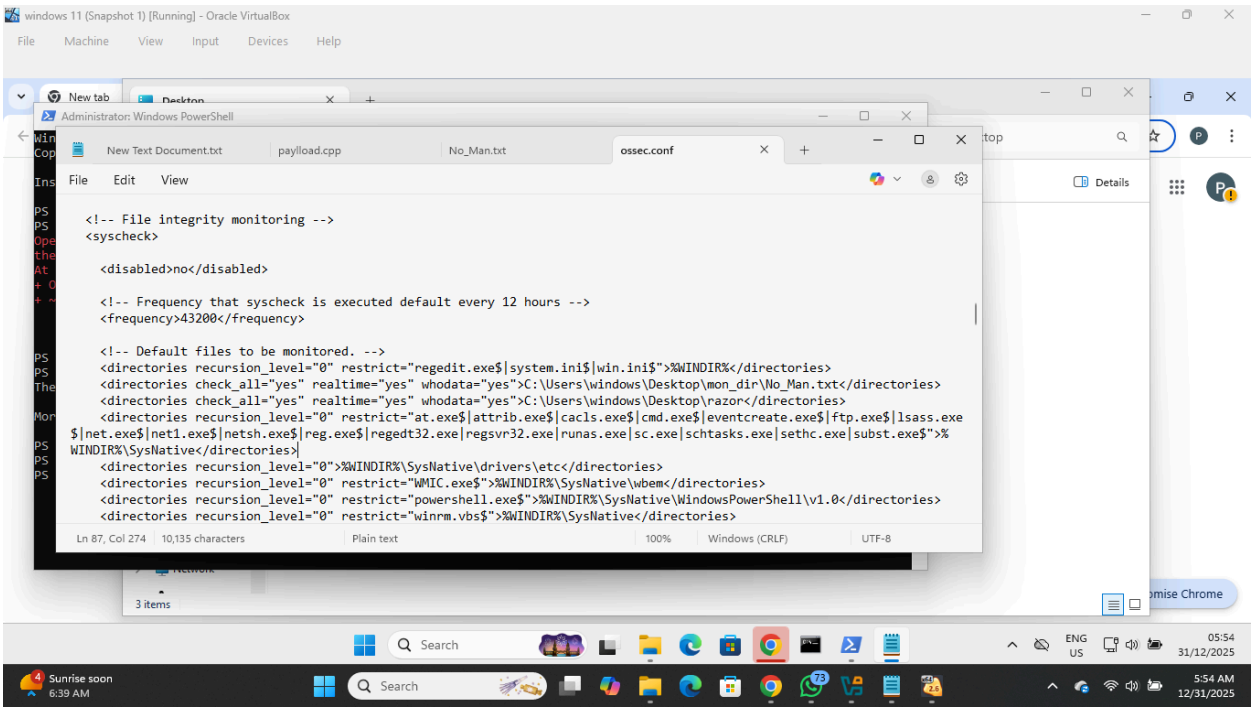
4. I modified the once empty file by adding one character to it



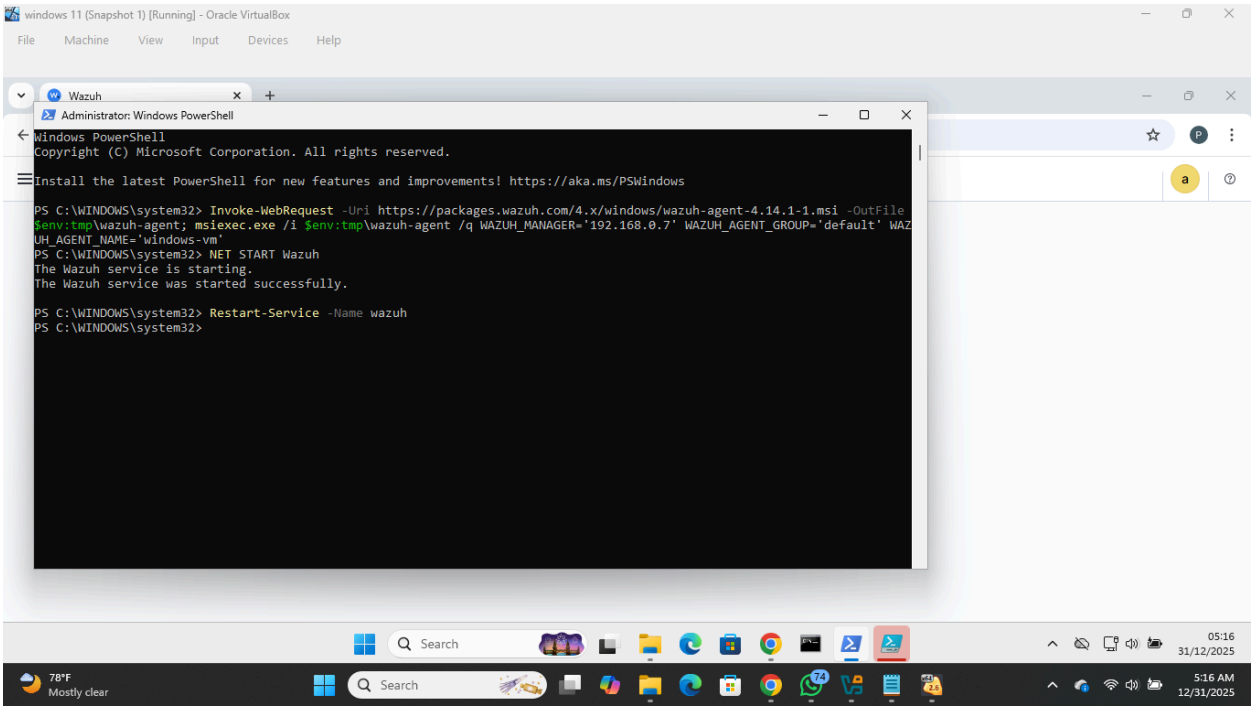
5. After modifying the file, I went back to my wazuh dashboard to confirm the event but the FIM recorded nothing.



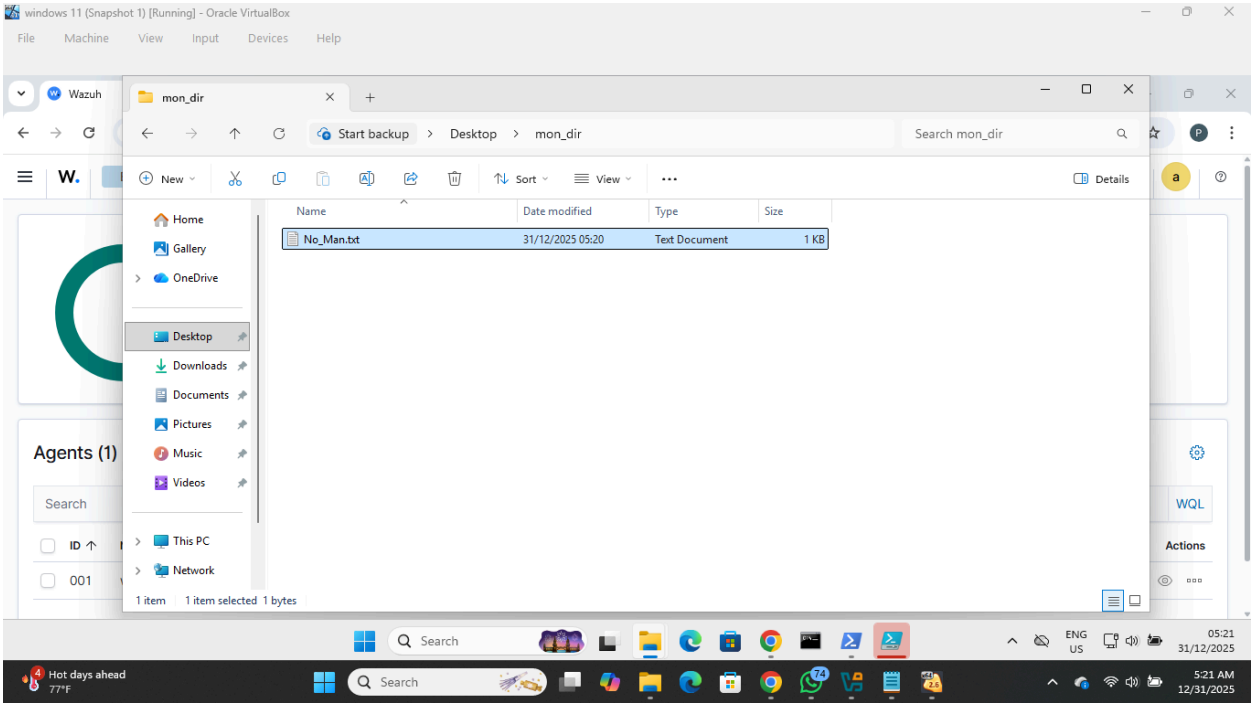
6. After troubleshooting and researching, I figured I had to also configure the "ossec.conf" file on my agent PC.
   On my agent PC, I configured the ossec.conf file at "C:\Program Files (x86)\ossec-agent\ossec.conf" using the exact same configuration as in the wazuh vm
   C:\Users\windows\Desktop\mon_dir\No_Man.txt
   C:\Users\windows\Desktop\razor

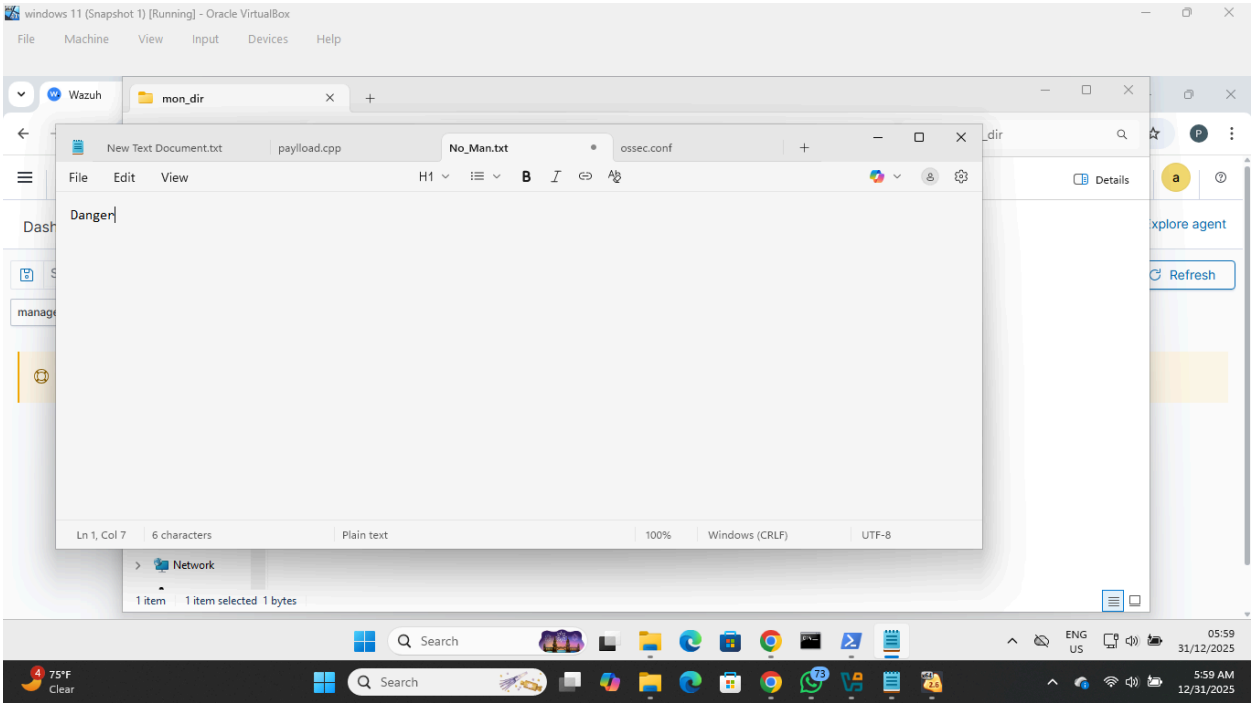


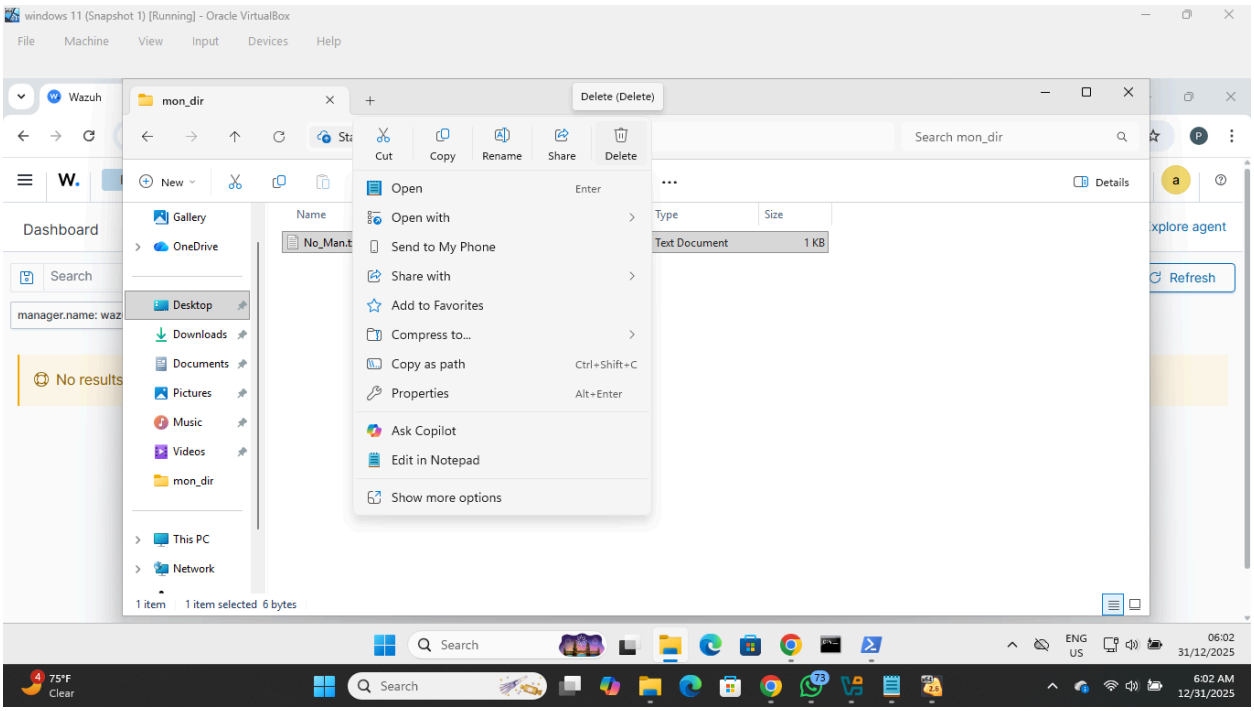
7. After configuring the agent “ossec.conf” file, I restarted the wazuh agent so the configuration would take effect.



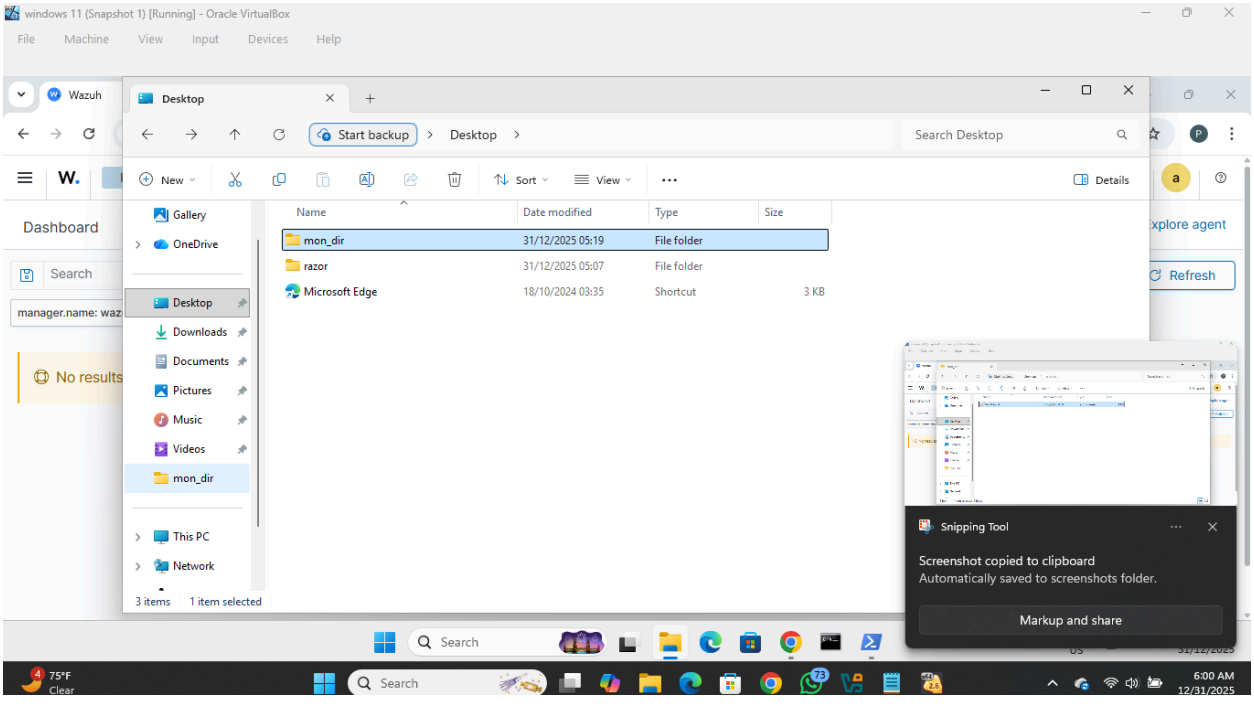
8. Next, I go back to testing the “No_Man.txt” file which is being monitored. I modified the file by adding text to it and saving the file. Then I deleted the file.

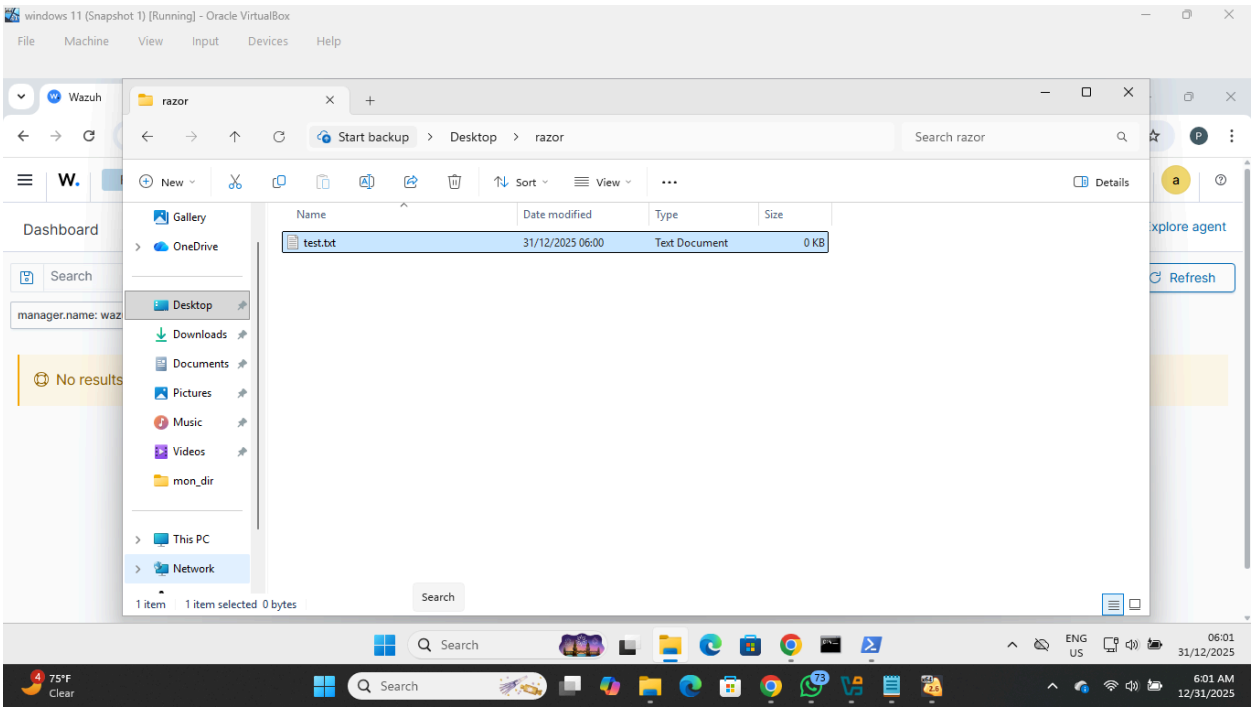


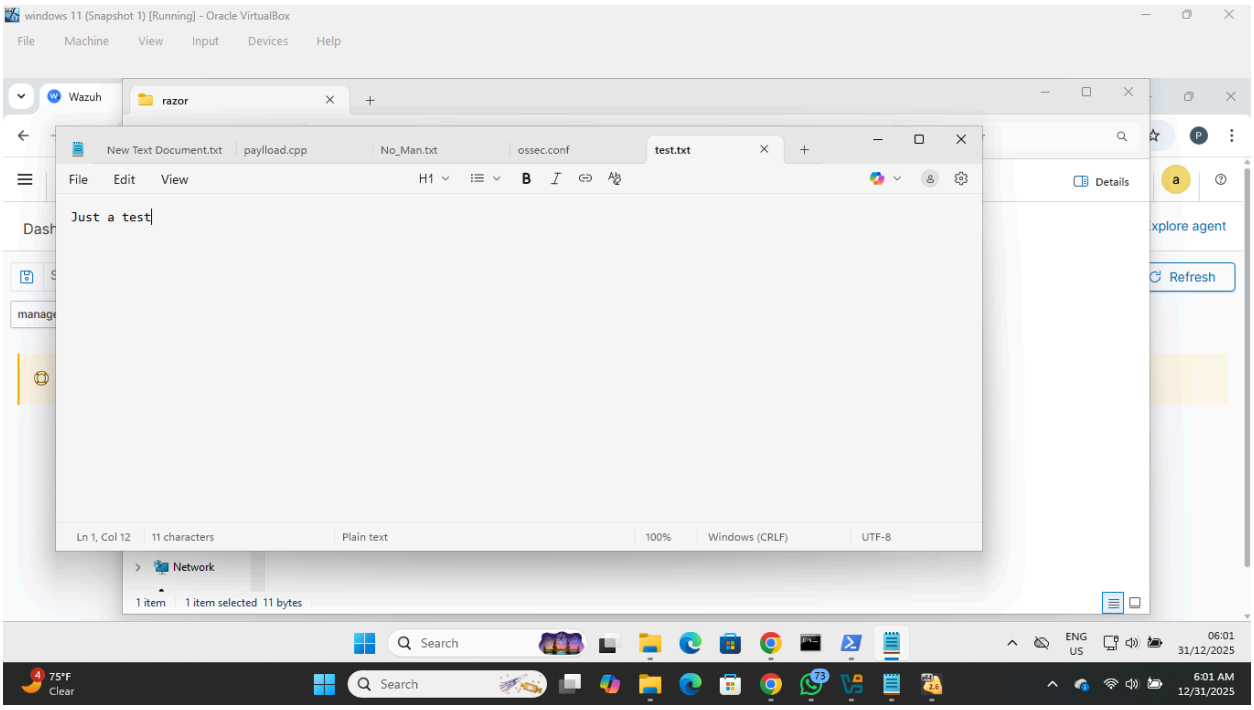


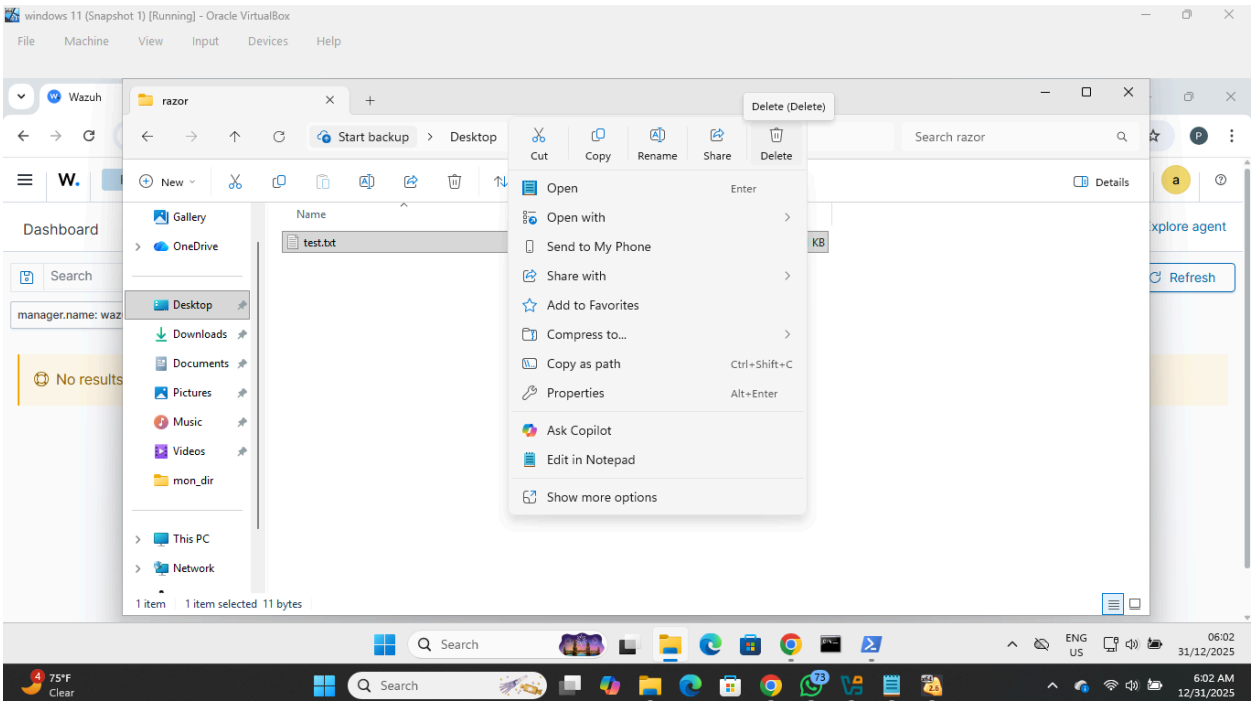


9. Next is the “mon_dir” directory which is also being monitored.
   In this directory, I created a new file named “test.txt”, modified it and deleted it.

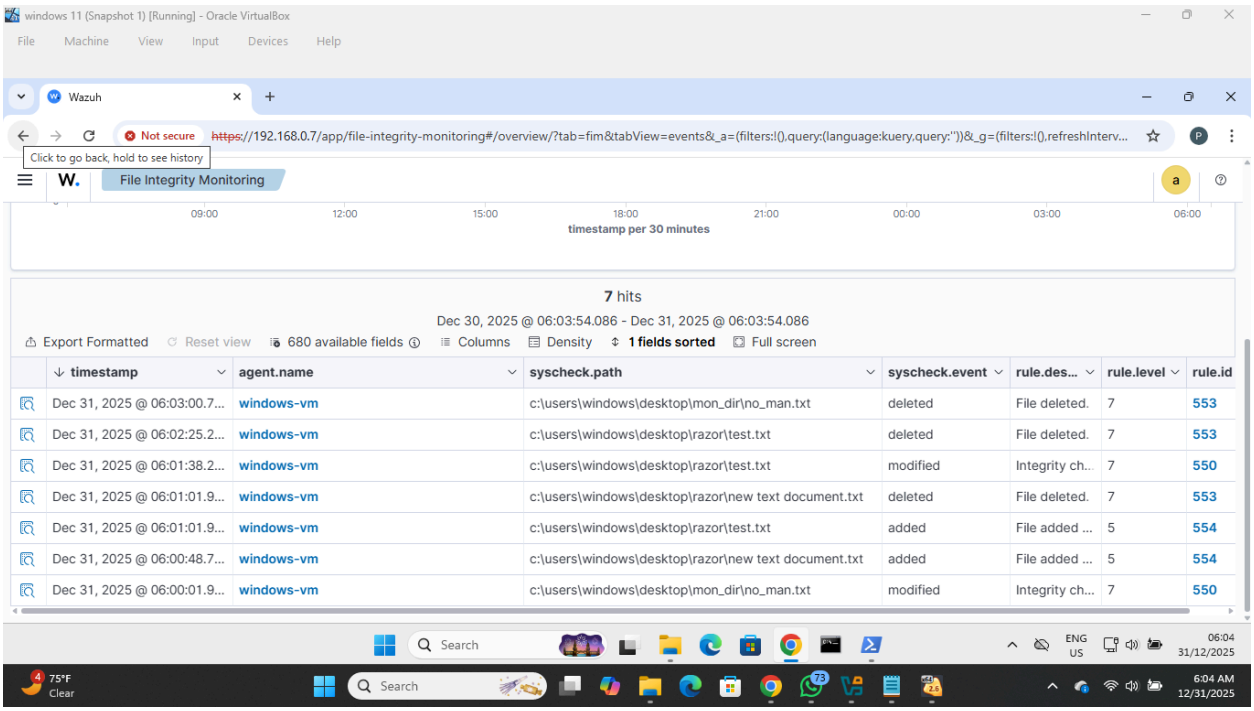


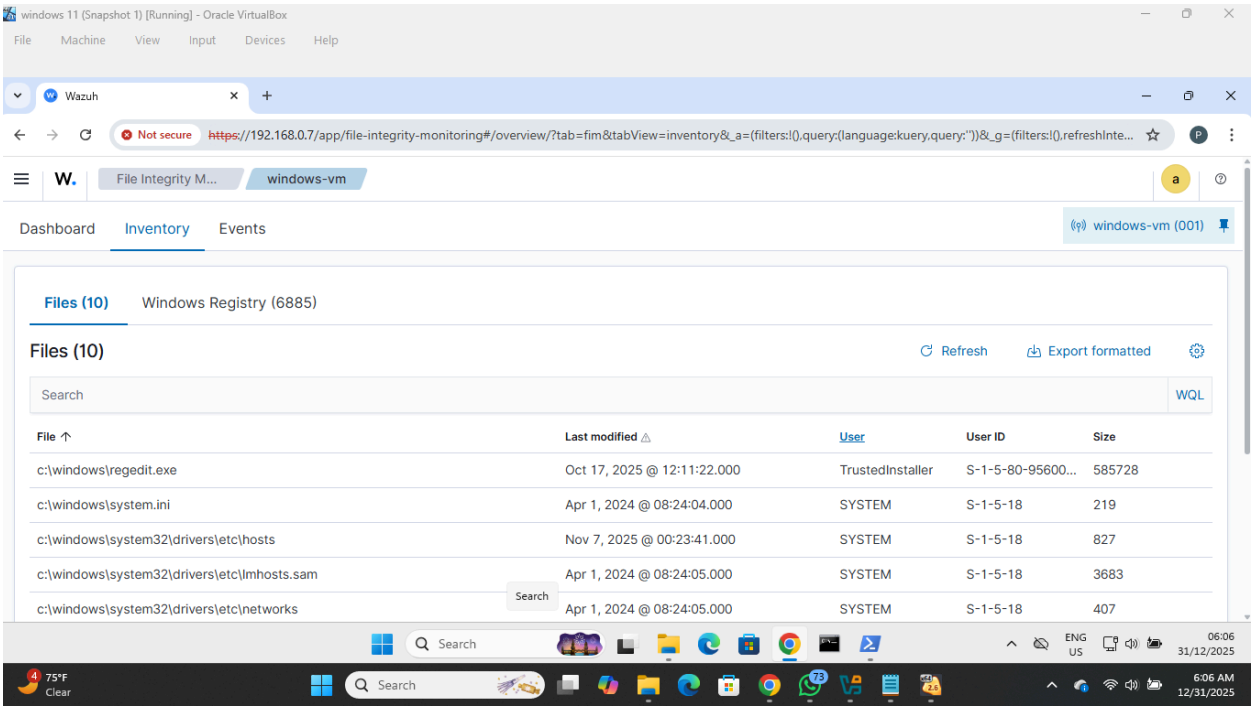

windows 11 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Wazuh
razor
Start backup › Desktop › razor
Search razor
New Sort View Details
Gallery
OneDrive
Desktop
Downloads
Documents
Pictures
Music
Videos
mon_dir
This PC
Network
Name Date modified Type Size
test.txt 31/12/2025 06:00 Text Document 0 KB
1 item 1 item selected 0 bytes
Dashboard
Search
Refresh
No results
Search
ENG US
06:01 31/12/2025
75°F Clear
6:01 AM 12/31/2025

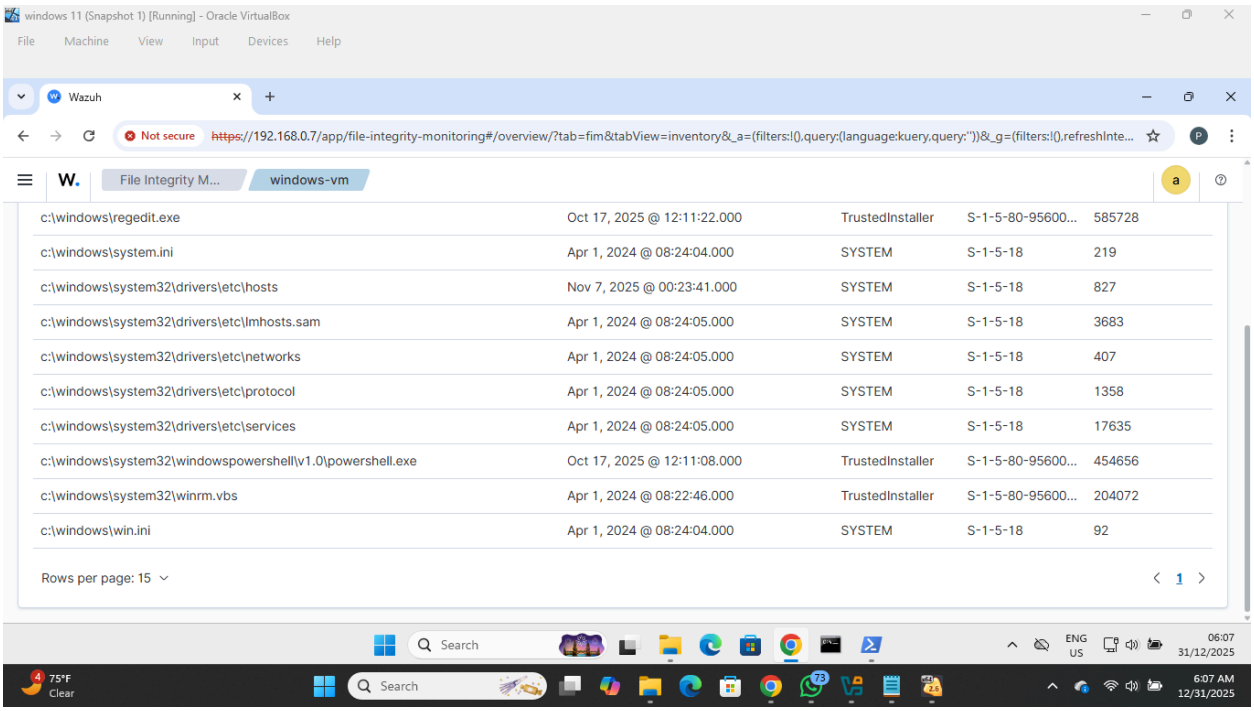

windows 11 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Wazuh
razor
New Text Document.txt
payload.cpp
No_Man.txt
ossec.conf
test.txt
File Edit View
Just a test
Ln 1, Col 12
11 characters
Plain text
100%
Windows (CRLF)
UTF-8
Network
1 item 1 item selected 11 bytes
Refresh
Search
ENG US
06:01 31/12/2025
75°F Clear
6:01 AM 12/31/2025

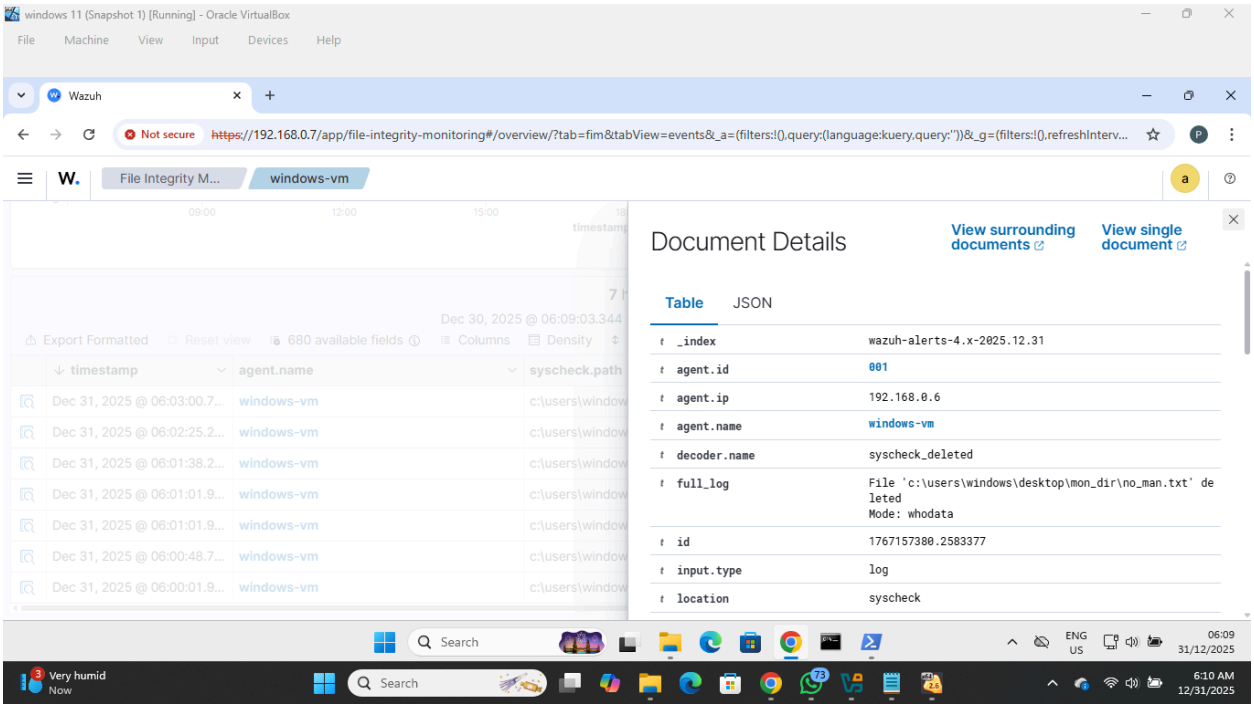


10. After performing the necessary modifications to the files and directories that are being monitored by my wazuh FIM, I went back to my wazuh dashboard to confirm.
    This time the wazuh FIM captured every event and modification.
    On the FIM events page, I can see every event along with their rule_ids.


windows 11 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Wazuh
Not secure https://192.168.0.7/app/file-integrity-monitoring#/overview/?tab=fim&tabView=events&_a=(filters:!(),query:(language:kuery,query:''))&_g=(filters:!(),refreshInterv...
Click to go back, hold to see history
File Integrity Monitoring
09:00 12:00 15:00 18:00 21:00 00:00 03:00 06:00
timestamp per 30 minutes
7 hits
Dec 30, 2025 @ 06:03:54.086 - Dec 31, 2025 @ 06:03:54.086
Export Formatted Reset view 680 available fields Columns Density 1 fields sorted Full screen
timestamp agent.name syscheck.path syscheck.event rule.des... rule.level rule.id
Dec 31, 2025 @ 06:03:00.7... windows-vm c:\users\windows\desktop\mon_dir\no_man.txt deleted File deleted. 7 553
Dec 31, 2025 @ 06:02:25.2... windows-vm c:\users\windows\desktop\razor\test.txt deleted File deleted. 7 553
Dec 31, 2025 @ 06:01:38.2... windows-vm c:\users\windows\desktop\razor\test.txt modified Integrity ch... 7 550
Dec 31, 2025 @ 06:01:01.9... windows-vm c:\users\windows\desktop\razor\new text document.txt deleted File deleted. 7 553
Dec 31, 2025 @ 06:01:01.9... windows-vm c:\users\windows\desktop\razor\test.txt added File added ... 5 554
Dec 31, 2025 @ 06:00:48.7... windows-vm c:\users\windows\desktop\razor\new text document.txt added File added ... 5 554
Dec 31, 2025 @ 06:00:01.9... windows-vm c:\users\windows\desktop\mon_dir\no_man.txt modified Integrity ch... 7 550
Search
ENG US
06:04 31/12/2025
75°F Clear
Search
6:04 AM 12/31/2025


windows 11 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Wazuh
Not secure https://192.168.0.7/app/file-integrity-monitoring#/overview/?tab=fim&tabView=inventory&_a=(filters:!(),query:(language:kuery,query:''))&_g=(filters:!(),refreshInte...
File Integrity M... windows-vm
Dashboard Inventory Events
windows-vm (001)
Files (10) Windows Registry (6885)
Files (10)
Refresh Export formatted
Search WQL
File Last modified User User ID Size
c:\windows\regedit.exe Oct 17, 2025 @ 12:11:22.000 TrustedInstaller S-1-5-80-95600... 585728
c:\windows\system.ini Apr 1, 2024 @ 08:24:04.000 SYSTEM S-1-5-18 219
c:\windows\system32\drivers\etc\hosts Nov 7, 2025 @ 00:23:41.000 SYSTEM S-1-5-18 827
c:\windows\system32\drivers\etc\lmhosts.sam Apr 1, 2024 @ 08:24:05.000 SYSTEM S-1-5-18 3683
c:\windows\system32\drivers\etc\networks Apr 1, 2024 @ 08:24:05.000 SYSTEM S-1-5-18 407
Search
Search
ENG US
06:06 31/12/2025
75°F Clear
Search
6:06 AM 12/31/2025


windows 11 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Wazuh
Not secure https://192.168.0.7/app/file-integrity-monitoring#/overview/?tab=fim&tabView=inventory&_a=(filters:!(),query:(language:kuery,query:''))&_g=(filters:!(),refreshInte...
W.
File Integrity M...
windows-vm
c:\windows\regedit.exe Oct 17, 2025 @ 12:11:22.000 TrustedInstaller S-1-5-80-95600... 585728
c:\windows\system.ini Apr 1, 2024 @ 08:24:04.000 SYSTEM S-1-5-18 219
c:\windows\system32\drivers\etc\hosts Nov 7, 2025 @ 00:23:41.000 SYSTEM S-1-5-18 827
c:\windows\system32\drivers\etc\lmhosts.sam Apr 1, 2024 @ 08:24:05.000 SYSTEM S-1-5-18 3683
c:\windows\system32\drivers\etc\networks Apr 1, 2024 @ 08:24:05.000 SYSTEM S-1-5-18 407
c:\windows\system32\drivers\etc\protocol Apr 1, 2024 @ 08:24:05.000 SYSTEM S-1-5-18 1358
c:\windows\system32\drivers\etc\services Apr 1, 2024 @ 08:24:05.000 SYSTEM S-1-5-18 17635
c:\windows\system32\windowspowershell\v1.0\powershell.exe Oct 17, 2025 @ 12:11:08.000 TrustedInstaller S-1-5-80-95600... 454656
c:\windows\system32\winrm.vbs Apr 1, 2024 @ 08:22:46.000 TrustedInstaller S-1-5-80-95600... 204072
c:\windows\win.ini Apr 1, 2024 @ 08:24:04.000 SYSTEM S-1-5-18 92
Rows per page: 15
1
Search
ENG US
06:07 31/12/2025
75°F Clear
Search
6:07 AM 12/31/2025


windows 11 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Wazuh
Not secure https://192.168.0.7/app/file-integrity-monitoring#/overview/?tab=fim&tabView=events&_a=(filters:!(),query:(language:kuery,query:''))&_g=(filters:!(),refreshInterv...
W.
File Integrity M...
windows-vm
Document Details
View surrounding documents
View single document
Table
JSON
_index wazuh-alerts-4.x-2025.12.31
agent.id 001
agent.ip 192.168.0.6
agent.name windows-vm
decoder.name syscheck_deleted
full_log File 'c:\users\windows\desktop\mon_dir\no_man.txt' deleted
Mode: whodata
id 1767157380.2583377
input.type log
location syscheck
Search
ENG US
06:09 31/12/2025
Very humid Now
Search
6:10 AM 12/31/2025

Checking GDPR & PCI DSS Compliance

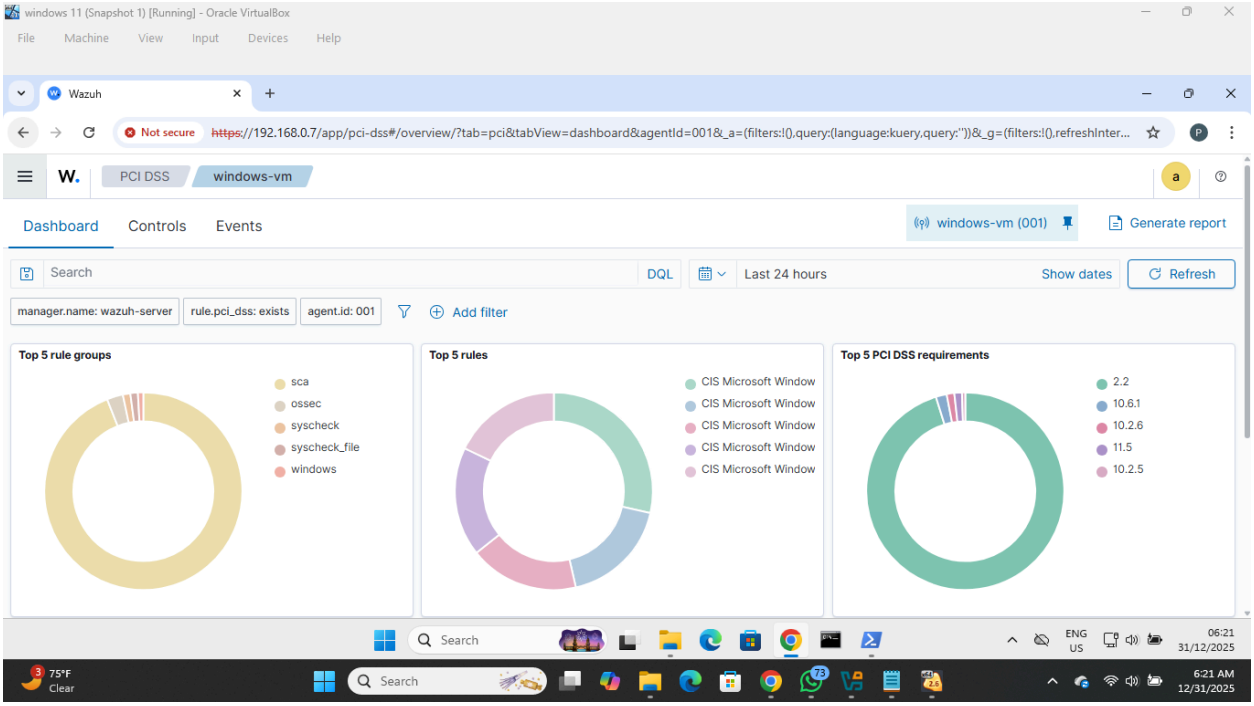
Wazuh automatically tags every alert with relevant compliance standards (like GDPR for data
privacy or PCI DSS for credit cards). This allows you to see if a security event violates a specific
law or regulation.

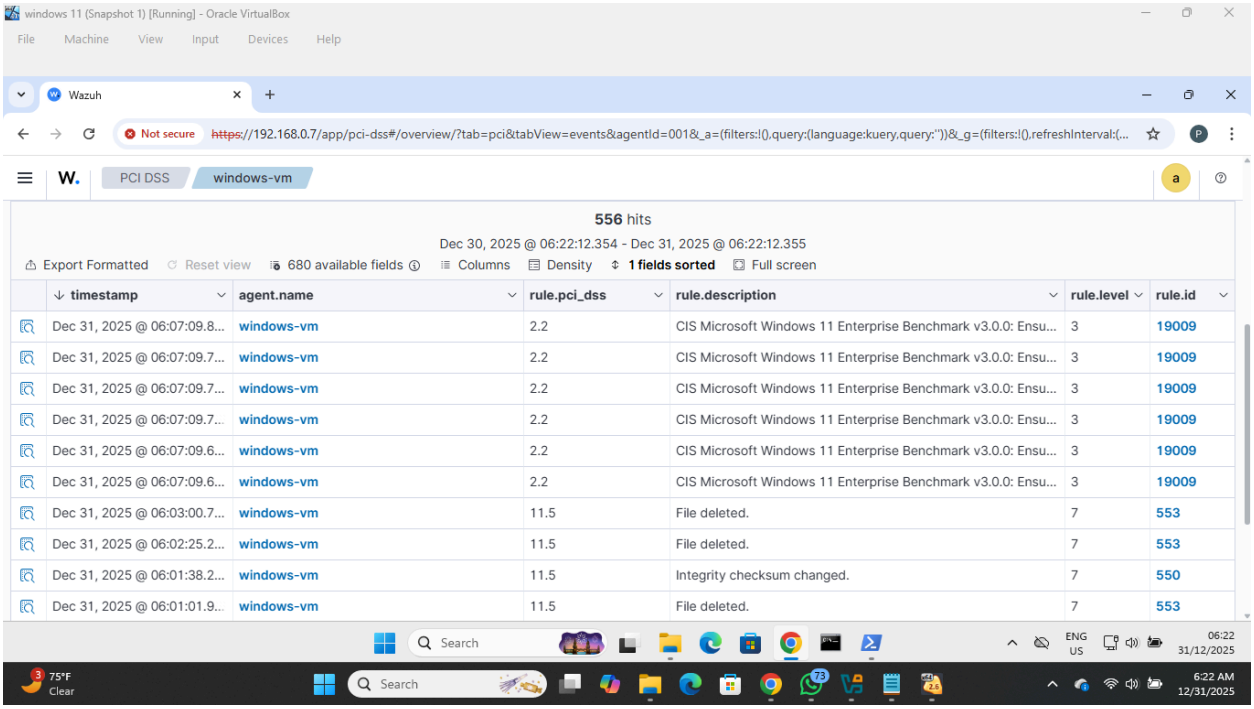
To check for events that violate the PCI_DSS and GDPR regulations, go to the wazuh menu bar, under the “security operations” section, you will find the PCI_DSS and GDPR options.
Select it to access events relating to PCI_DSS or GDPR

We can also filter for events relating to GDPR and PCI_DSS under the security events using filters.

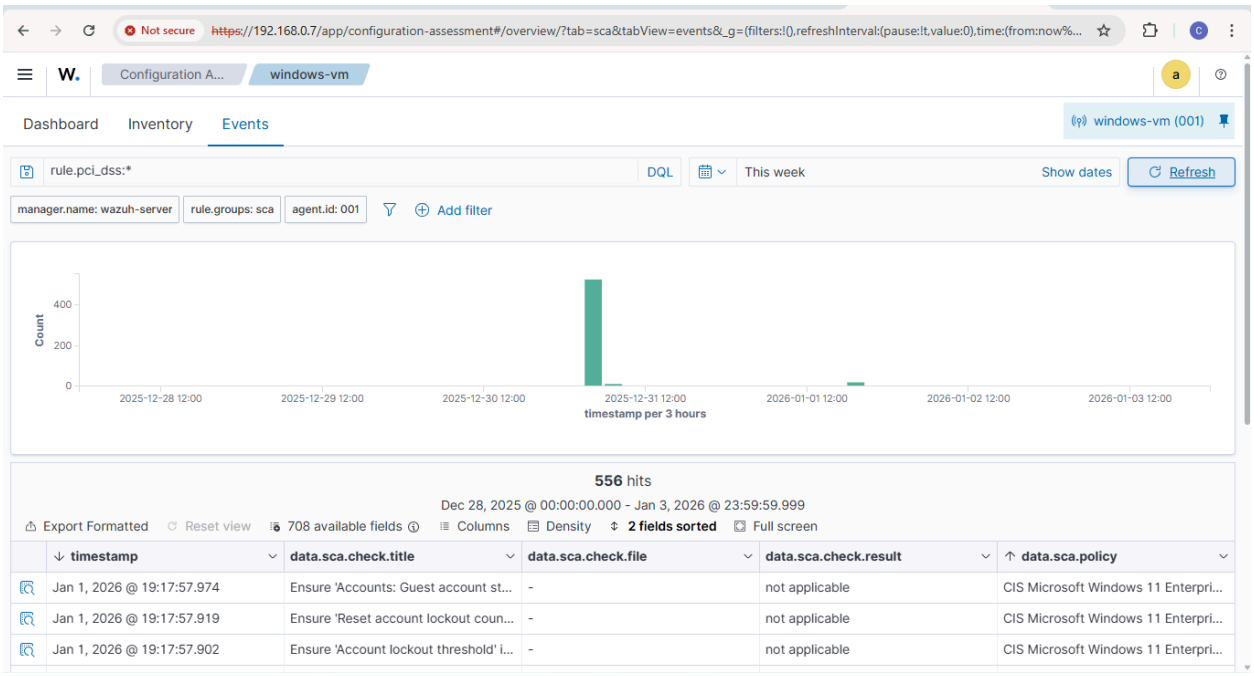
For PCI DSS: Type rule.pci_dss: * and press Enter

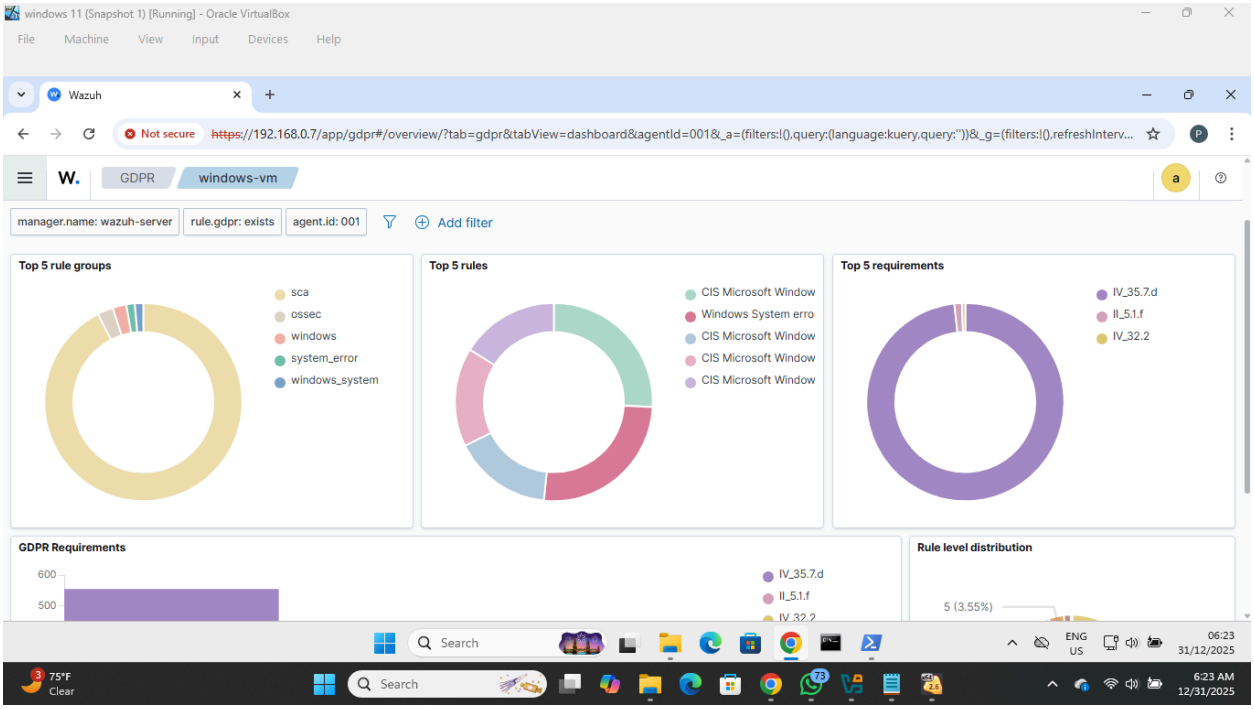
For GDPR: Type rule.gdpr: * and press Enter.

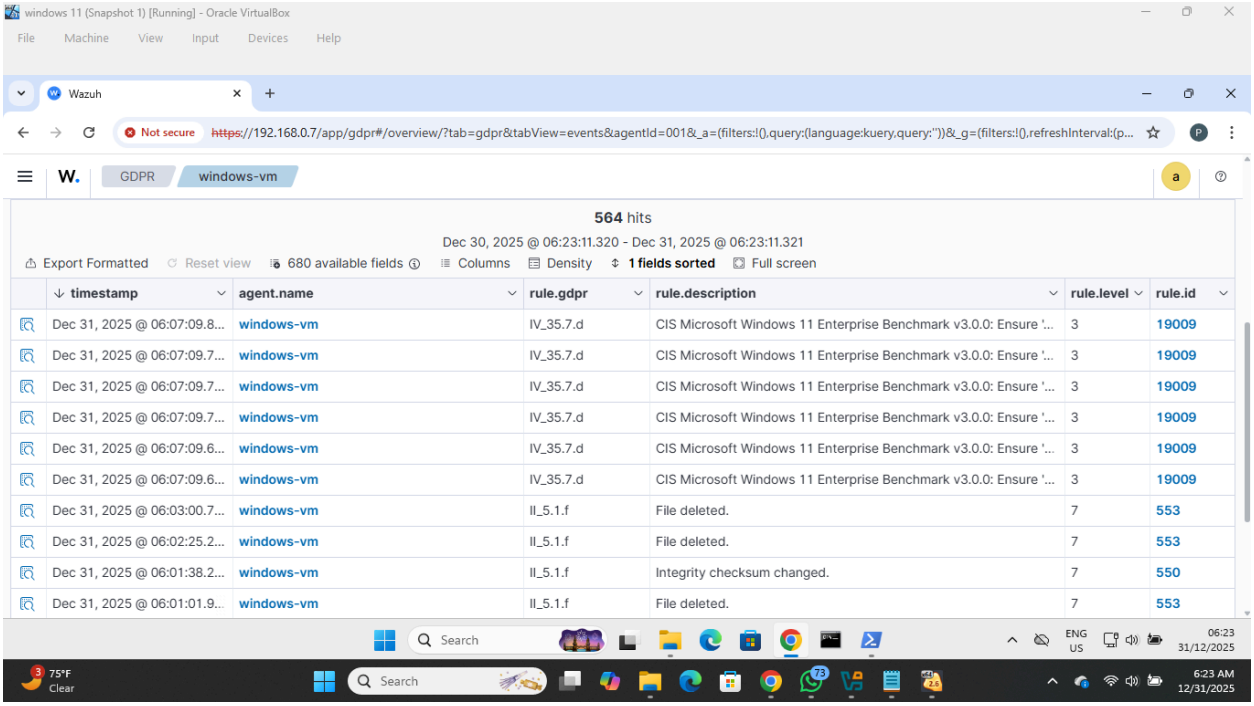




For PCI DSS: Type rule.pci_dss: * and press Enter




windows 11 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Wazuh
Not secure https://192.168.0.7/app/gdpr#/overview/?tab=gdpr&tabView=dashboard&agentId=001&_a=(filters:!(),query:(language:kuery,query:''))&_g=(filters:!(),refreshInterv...
GDPR
windows-vm
manager.name: wazuh-server
rule.gdpr: exists
agent.id: 001
Add filter
Top 5 rule groups
sca
ossec
windows
system_error
windows_system
Top 5 rules
CIS Microsoft Window
Windows System erro
CIS Microsoft Window
CIS Microsoft Window
CIS Microsoft Window
Top 5 requirements
IV_35.7.d
II_5.1.f
IV_32.2
GDPR Requirements
600
500
IV_35.7.d
II_5.1.f
IV_32.2
Rule level distribution
5 (3.55%)
Search
ENG US
06:23 31/12/2025
75°F Clear
Search
6:23 AM 12/31/2025


windows 11 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Wazuh
Not secure https://192.168.0.7/app/gdpr#/overview/?tab=gdpr&tabView=events&agentId=001&_a=(filters:!(),query:(language:kuery,query:''))&_g=(filters:!(),refreshInterval:(p...
GDPR
windows-vm
564 hits
Dec 30, 2025 @ 06:23:11.320 - Dec 31, 2025 @ 06:23:11.321
Export Formatted
Reset view
680 available fields
Columns
Density
1 fields sorted
Full screen
timestamp
agent.name
rule.gdpr
rule.description
rule.level
rule.id
Dec 31, 2025 @ 06:07:09.8... windows-vm IV_35.7.d CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Ensure '... 3 19009
Dec 31, 2025 @ 06:07:09.7... windows-vm IV_35.7.d CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Ensure '... 3 19009
Dec 31, 2025 @ 06:07:09.7... windows-vm IV_35.7.d CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Ensure '... 3 19009
Dec 31, 2025 @ 06:07:09.7... windows-vm IV_35.7.d CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Ensure '... 3 19009
Dec 31, 2025 @ 06:07:09.6... windows-vm IV_35.7.d CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Ensure '... 3 19009
Dec 31, 2025 @ 06:07:09.6... windows-vm IV_35.7.d CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Ensure '... 3 19009
Dec 31, 2025 @ 06:03:00.7... windows-vm II_5.1.f File deleted. 7 553
Dec 31, 2025 @ 06:02:25.2... windows-vm II_5.1.f File deleted. 7 553
Dec 31, 2025 @ 06:01:38.2... windows-vm II_5.1.f Integrity checksum changed. 7 550
Dec 31, 2025 @ 06:01:01.9... windows-vm II_5.1.f File deleted. 7 553
Search
ENG US
06:23 31/12/2025
75°F Clear
Search
6:23 AM 12/31/2025

For GDPR: Type rule.gdpr: * and press Enter.

